

Unit 5

- **Optimization**
- **Modern tools**
- **Genetic Algorithm**
- **AI & ML**
- **Data Analytics**
- **Artificial Neural Network**

Optimization

Find values of the variables that minimize or maximize the objective function while satisfying the constraints.

Components of Optimization Problem

- **Objective Function:** An objective function which we want to minimize or maximize.
- For example, in a manufacturing process, we might want to maximize the profit or minimize the cost.
- In fitting experimental data to a user-defined model, we might minimize the total deviation of observed data from predictions based on the model.
- In designing an inductor, we might want to maximize the Quality Factor and minimize the area.

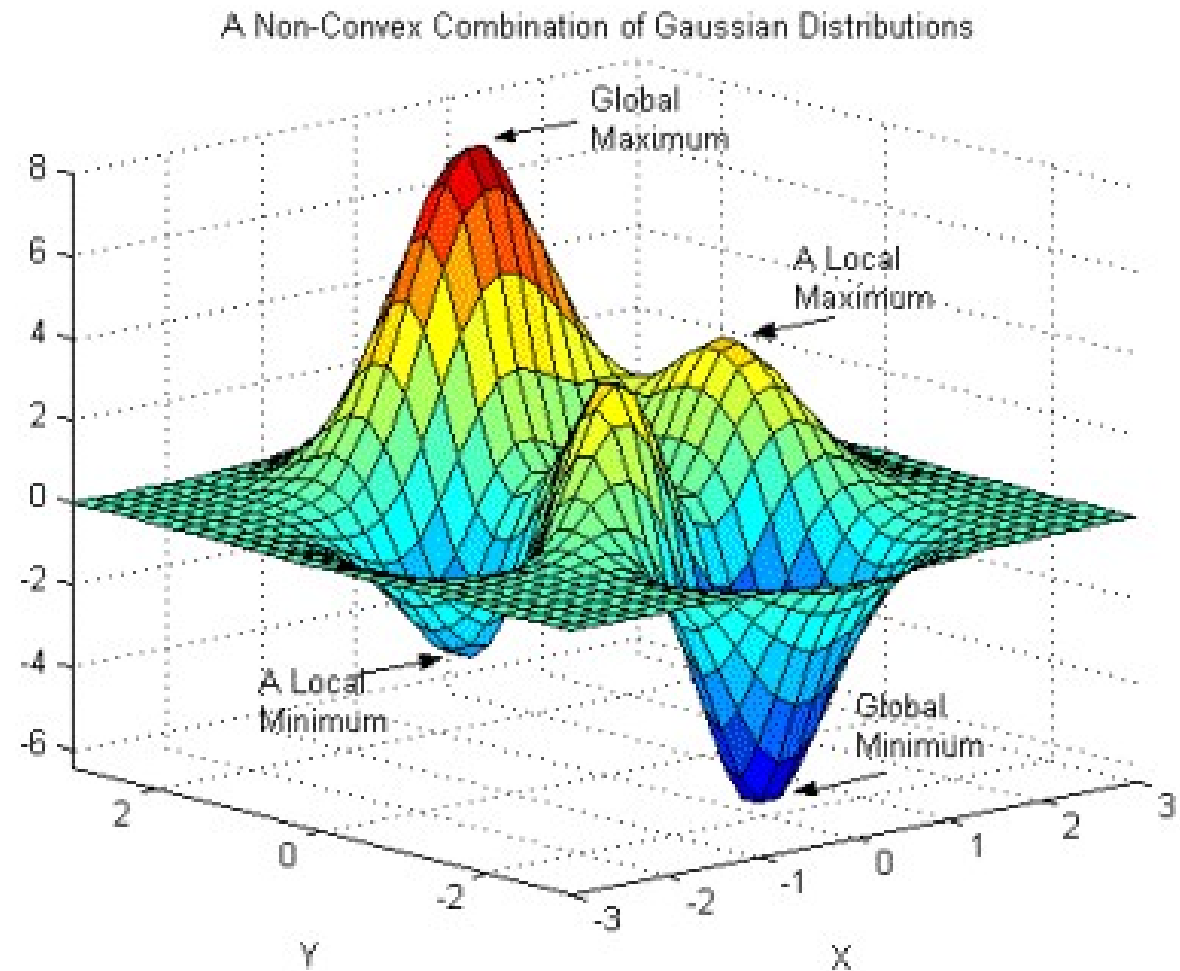
Component of Optimization Problem

- **Design Variables:** A set of unknowns or variables which affect the value of the objective function.
- In the manufacturing problem, the variables might include the amounts of different resources used or the time spent on each activity.
- In fitting-the-data problem, the unknowns are the parameters that define the model.
- In the inductor design problem, the variables used define the layout geometry of the panel.

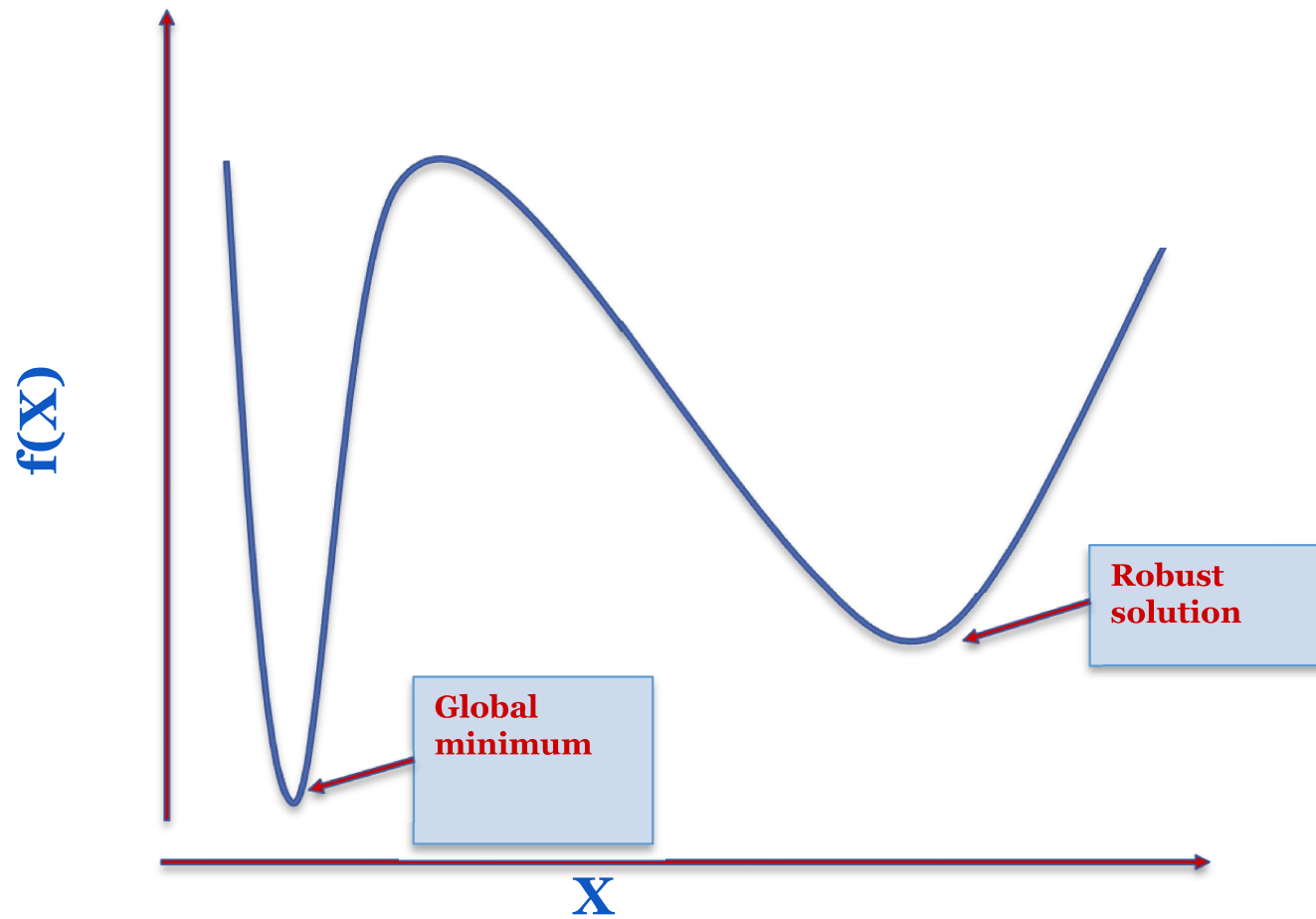
Component of Optimization Problem

- **Constraints:** A set of constraints that allow the unknowns to take on certain values but exclude others.
- For the manufacturing problem, it does not make sense to spend a negative amount of time on any activity, so we constrain all the "time" variables to be non-negative.
- In the inductor design problem, we would probably want to limit the upper and lower value of layout parameters and to target an inductance value within the tolerance level.

- ✓ A **local optimum** of a function is a point where the function value is smaller/larger than or equal to the value at nearby points, but possibly greater than at a distant point.
- ✓ A **global optimum** is a point where the function value is smaller/larger than or equal to the value at all other feasible points.

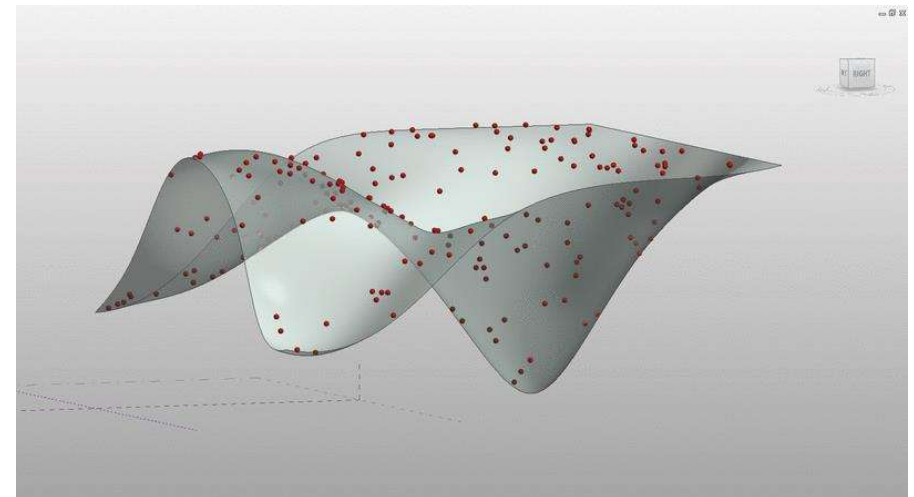
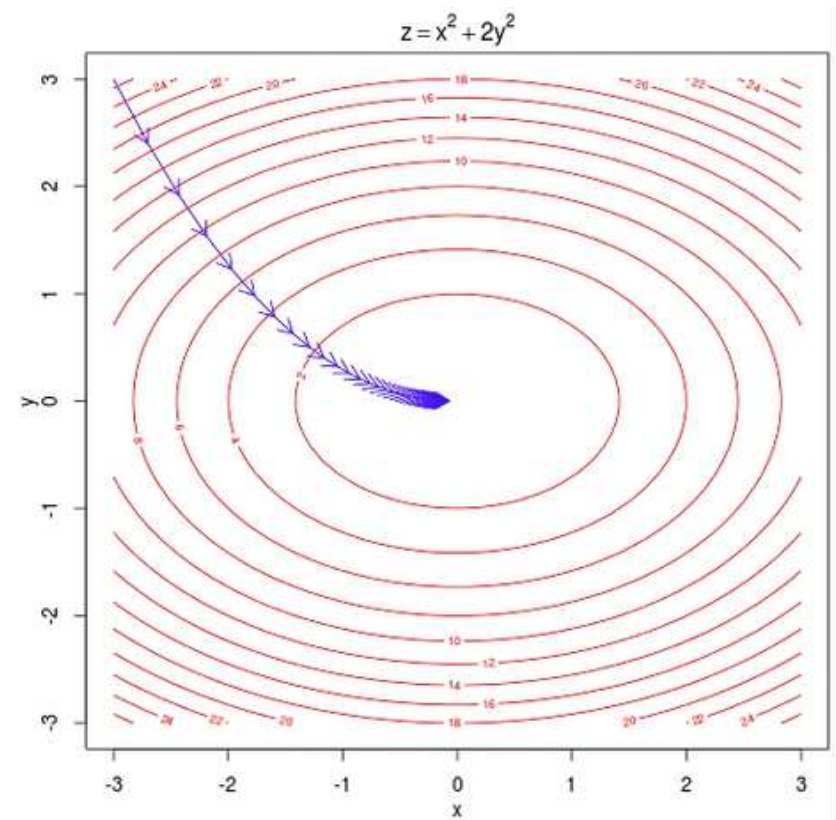


Robust Solution



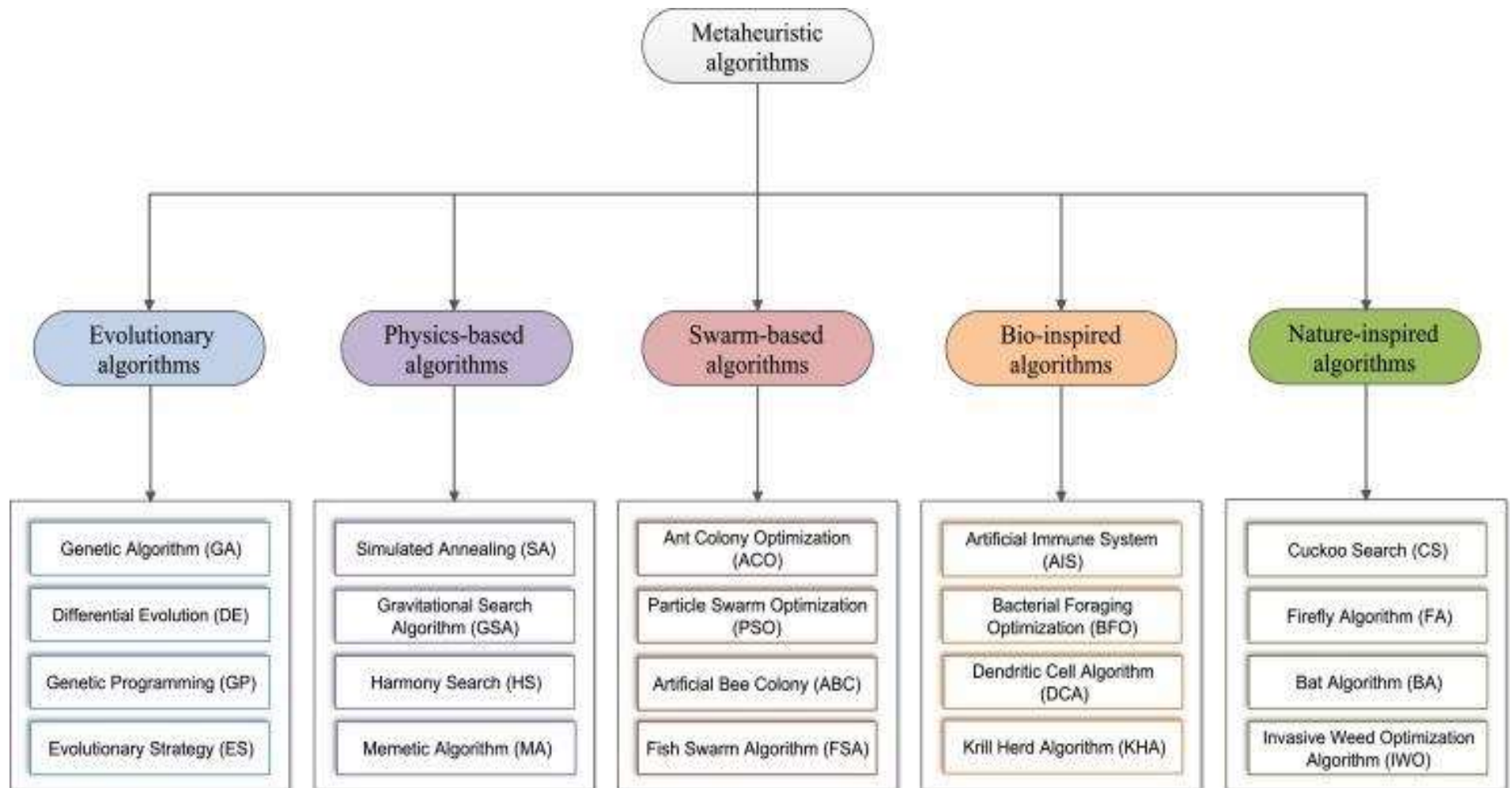
**Derivative-based and Classical
optimization**

**Population-based (metaheuristic)
optimization**



Derivative free Metaheuristic Algorithm

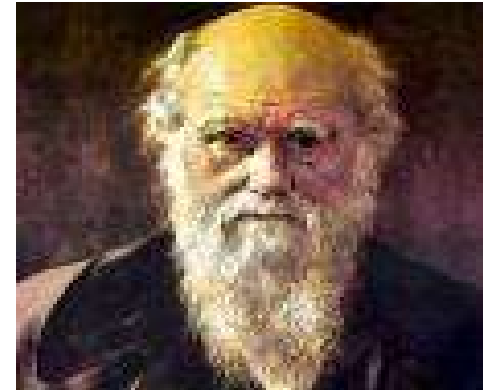
- ❑ Algorithms with stochastic components were often referred to as heuristic in the past, though the recent literature tends to refer to them as metaheuristics.
- ❑ Heuristic means to find or to discover by trial and error.
Here meta- means beyond or higher level, and metaheuristics generally perform better than simple heuristics.
- ❑ A metaheuristic can be considered as a master strategy that guides and modifies other heuristics to produce solutions beyond those that are normally generated in a quest for local optimality.
- ❑ All metaheuristic algorithms use a certain tradeoff of randomization and local search.



A few modern tools for optimization

Genetic algorithm

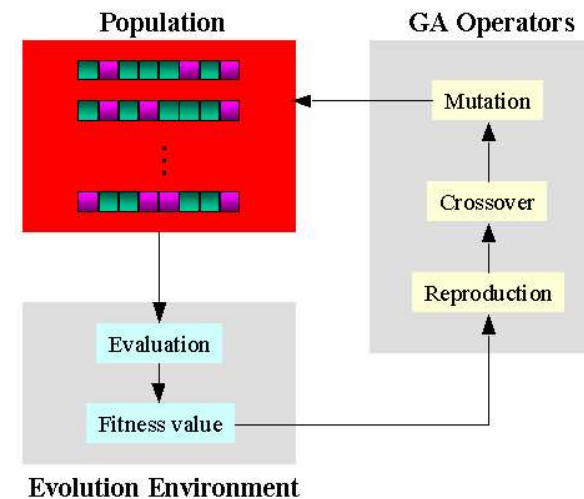
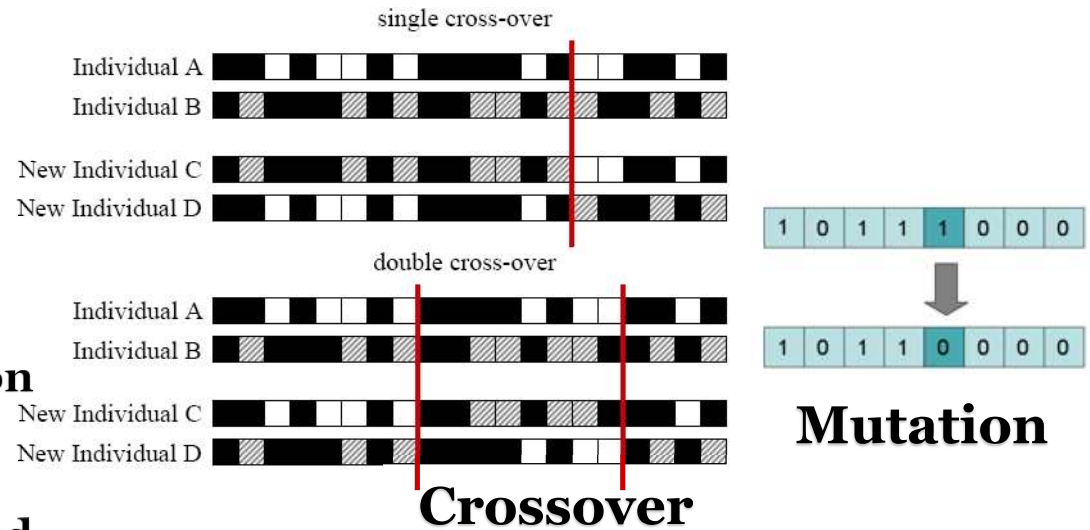
Algorithms that mimics natural genetics and natural selection i.e. the theory of “Survival of the fittest”



The basic concept of GA is designed to simulate processes in natural system necessary for evolution, specifically those that follow the principles first laid down by Charles Darwin. As such they represent an intelligent exploitation of a random search within a defined search space to solve a problem. .

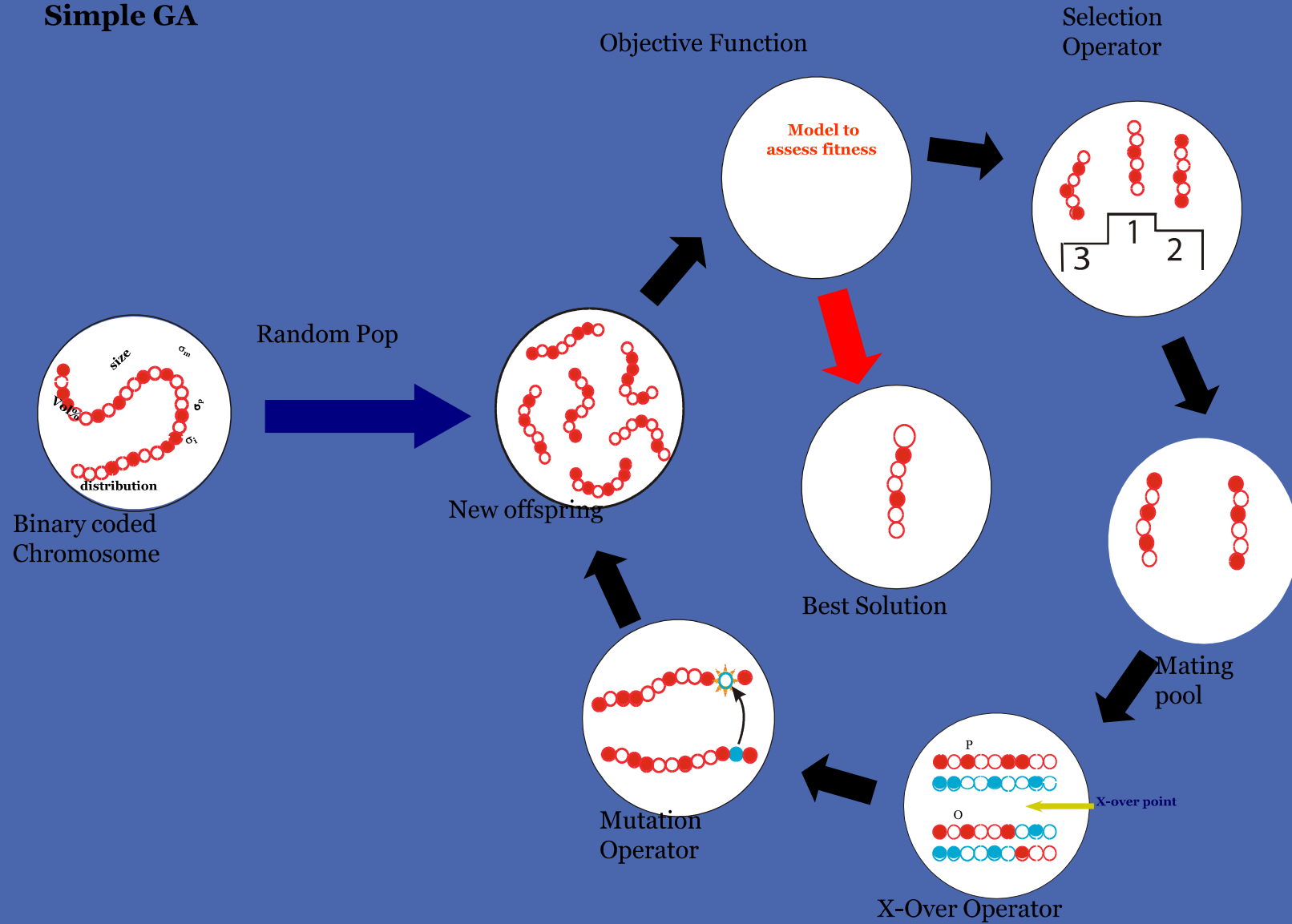
Stages of Genetic algorithm

- Chromosome formation
- Random population creation
- Fitness assessment
- Mating between the survived section
- Crossover
- Mutation
- New offspring creation

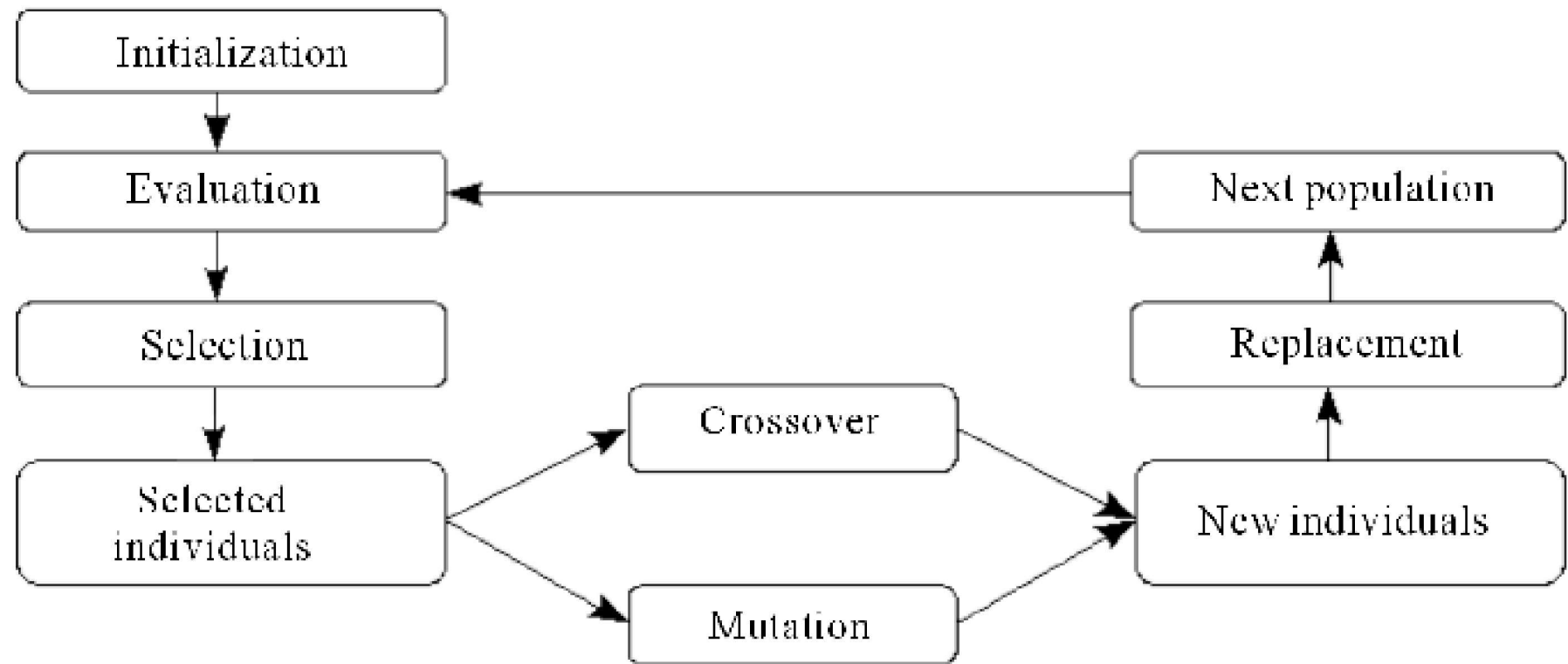


Genetic Algorithm Evolution Flow

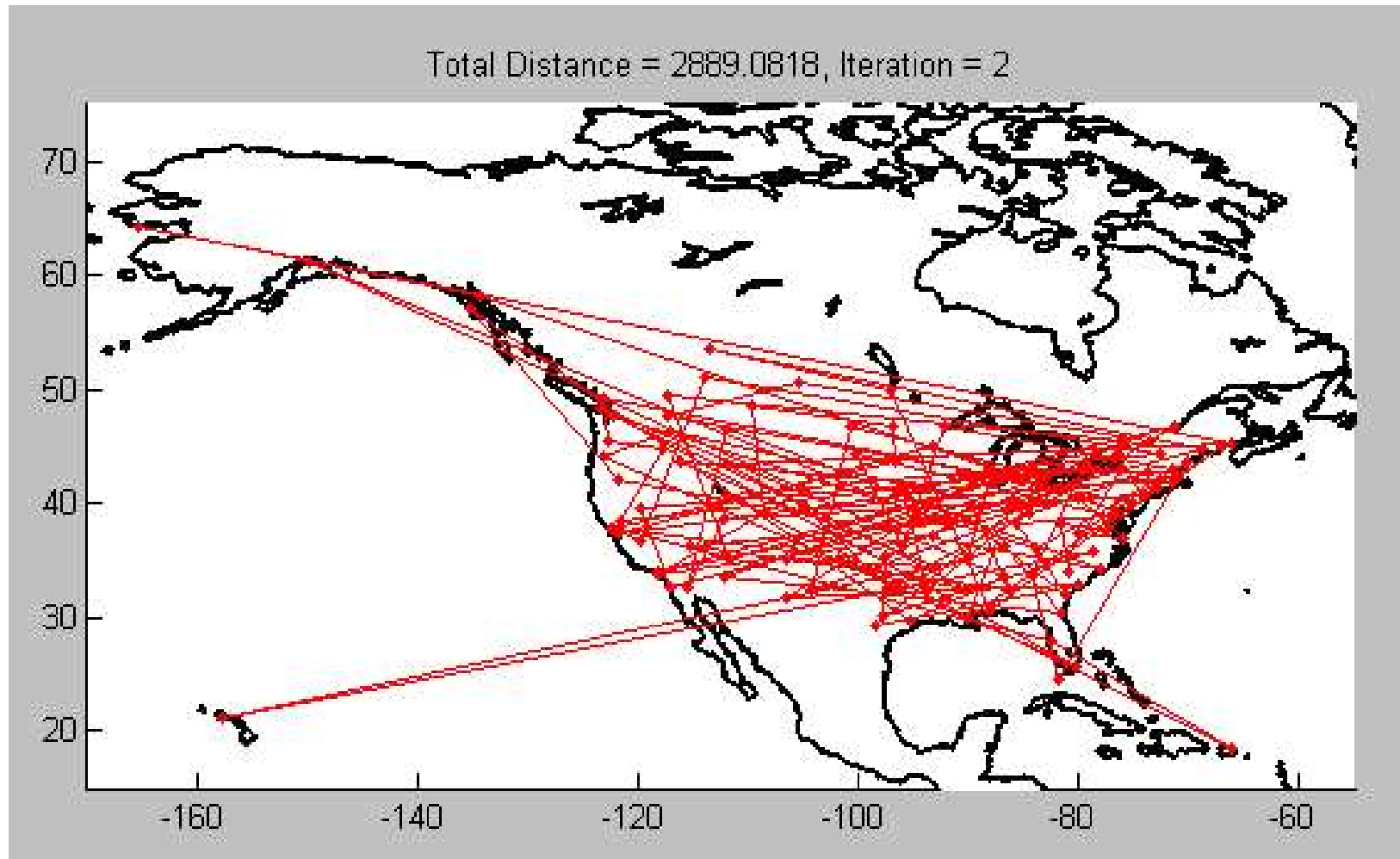
Simple GA



Scheme of Genetic algorithm



Travelling salesman problem



Simulated Annealing

Annealing

- Annealing is a thermal process for obtaining low energy states of a solid in a heat bath.
- The process contains two steps:
 - Increase the temperature of the heat bath to a maximum value at which the solid melts.
 - Decrease carefully the temperature of the heat bath until the particles arrange themselves in the ground state of the solid.
Ground state is a minimum energy state of the solid.
- The ground state of the solid is obtained only if the maximum temperature is high enough and the cooling is done slowly.

Simulated Annealing

- The process of annealing can be simulated with the Metropolis algorithm, which is based on Monte Carlo techniques.
- We can apply this algorithm to generate a solution to combinatorial optimization problems assuming an analogy between them and physical many-particle systems with the following equivalences:
 - Solutions in the problem are equivalent to states in a physical system.
 - The cost of a solution is equivalent to the “energy” of a state.

Simulated Annealing

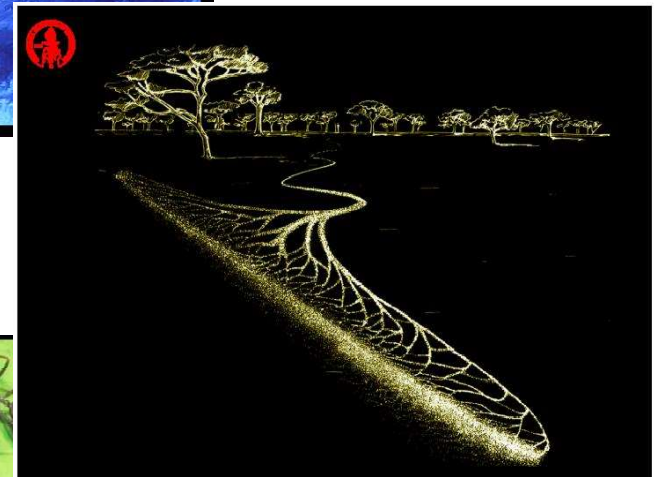
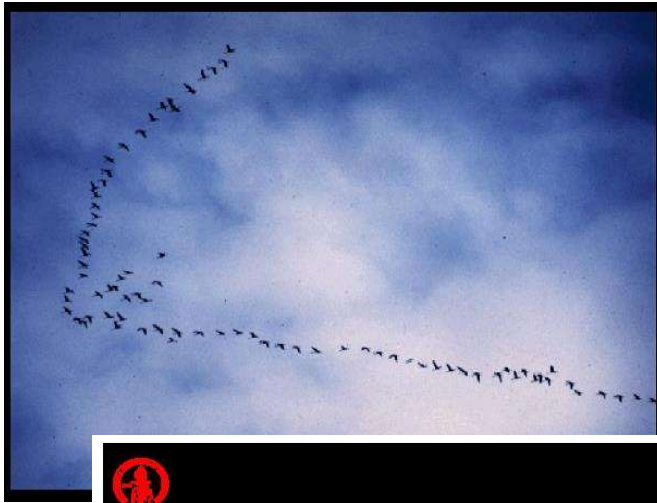
- To apply simulated annealing with optimization purposes we require the following:
 - A successor function that returns a “close” neighboring solution given the actual one. This will work as the “disturbance” for the particles of the system.
 - A target function to optimize that depends on the current state of the system. This function will work as the energy of the system.
- The search is started with a randomized state. In a polling loop we will move to neighboring states always accepting the moves that decrease the energy while only accepting bad moves accordingly to a probability distribution dependent on the “temperature” of the system.

Simulated Annealing

- Decrease the temperature slowly, accepting less bad moves at each temperature level until at very low temperatures the algorithm becomes a greedy hill-climbing algorithm.
 - The distribution used to decide if we accept a bad movement is known as Boltzmann distribution.
- This distribution is very well known in solid physics and plays a central role in simulated annealing. Where y is the current configuration of the system, E_y is the energy related with it, and Z is a normalization constant.

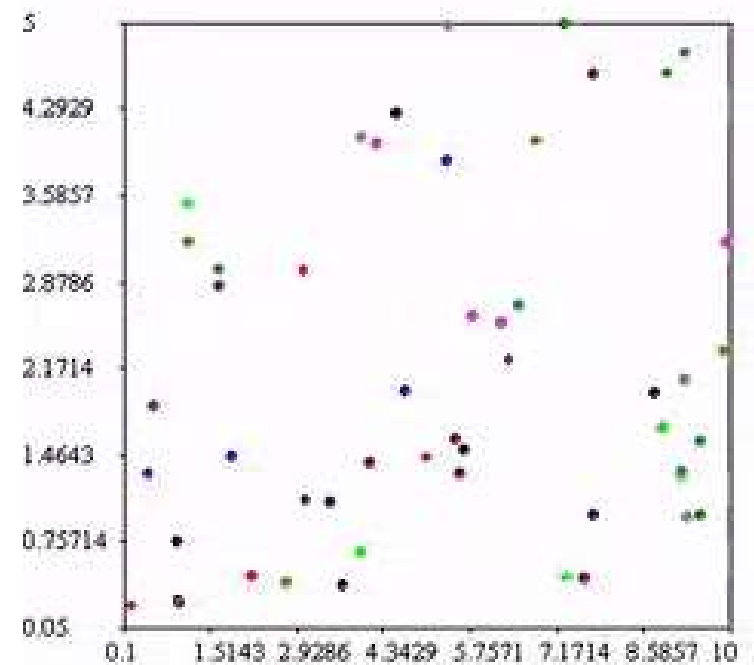
Swarm intelligence

- Collective system capable of accomplishing difficult tasks in dynamic and varied environments without any external guidance or control and with no central coordination
- Achieving a collective performance which could not normally be achieved by an individual acting alone
- Constituting a natural model particularly suited to distributed problem solving



Particle Swarm Optimization

- Metaheuristic technique based on the movement and intelligence of swarms looking for the most fertile feeding location
- Simple algorithm, easy to implement and few parameters to adjust mainly the velocity
- A “swarm” is an apparently disorganized collection (population) of moving individuals that tend to cluster together while each individual seems to be moving in a random direction.



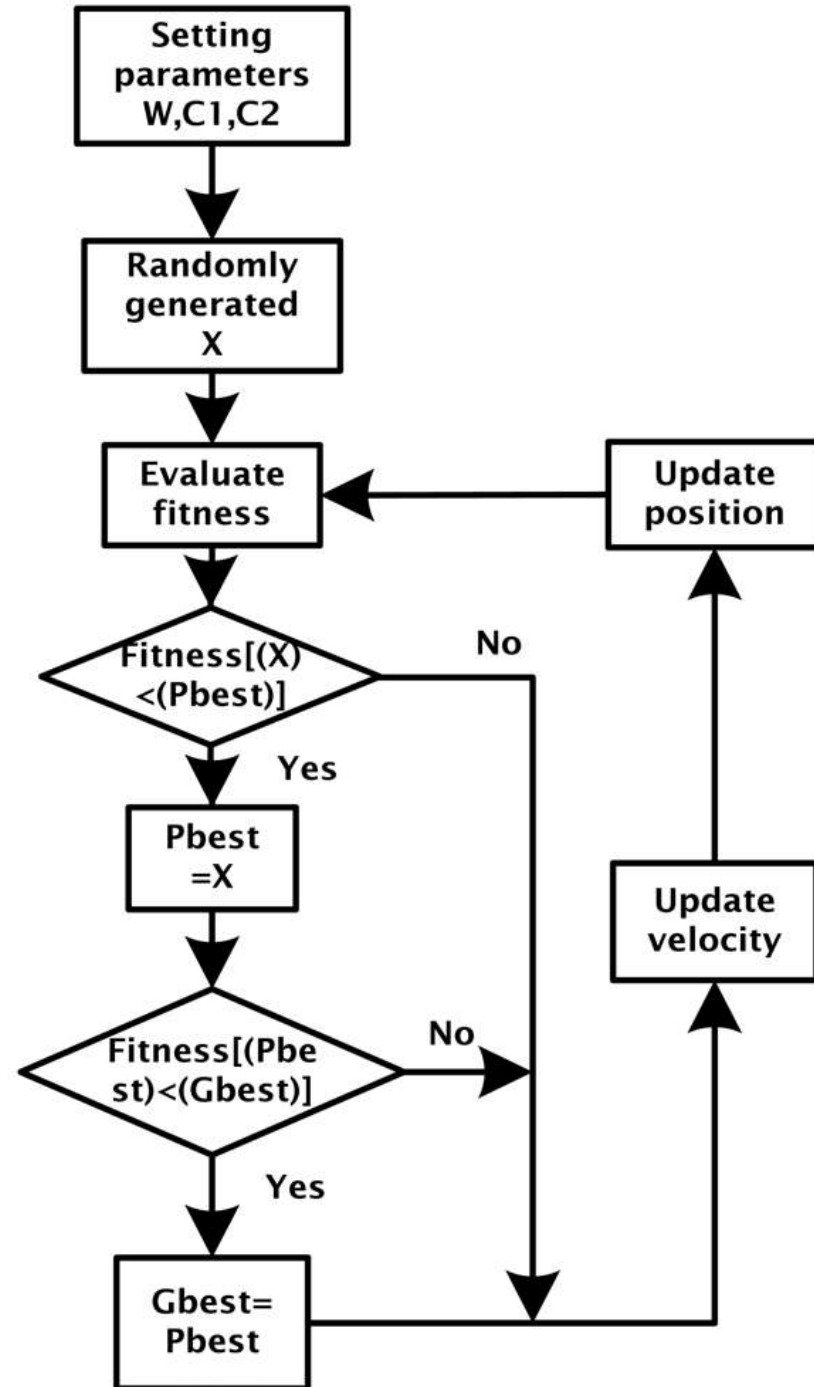
Particle Swarm Optimization

- It uses a number of agents (particles) that constitute a swarm moving around in the search space looking for the best solution.
- Each particle is treated as a point in a D-dimensional space which adjusts its “flying” according to its own flying experience as well as the flying experience of other particles
- Each particle keeps track of its coordinates in the problem space which are associated with the best solution (fitness) that has achieved so far. This value is called ***pbest***.

Particle Swarm Optimization

- Another best value that is tracked by the PSO is the best value obtained so far by any particle in the neighbors of the particle. This value is called ***gbest***.
- The PSO concept consists of changing the velocity (or acceleration) of each particle toward its *pbest* and the *gbest* position at each time step.

Scheme of Particle Swarm Optimization

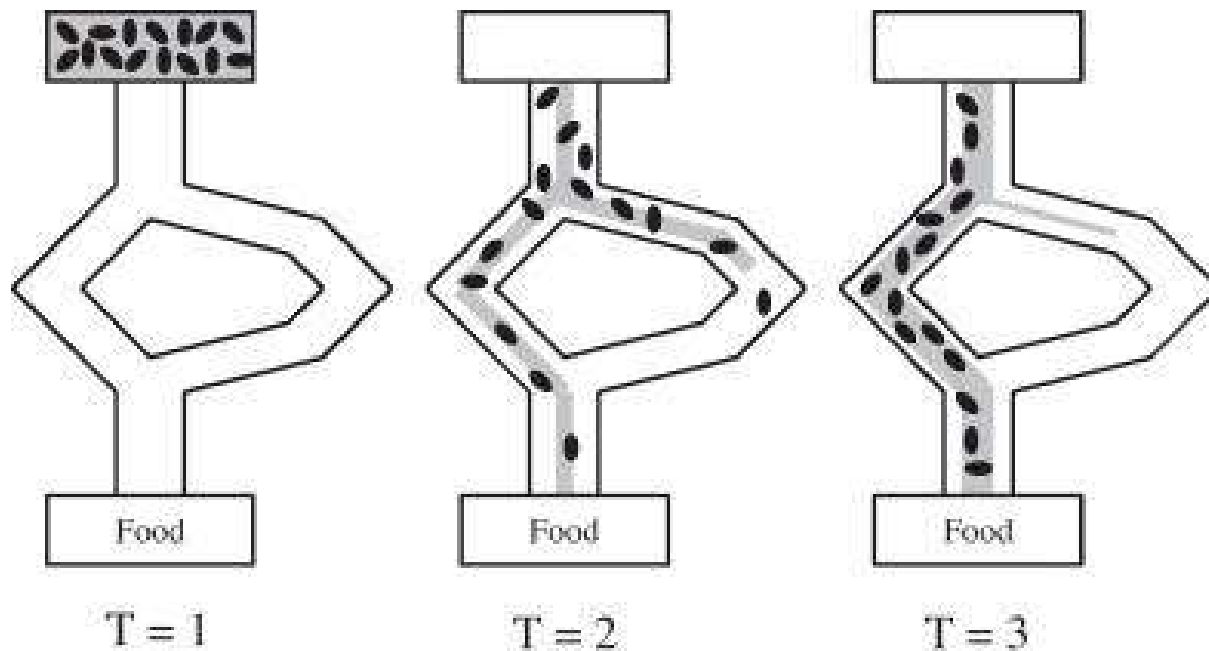


Ant Colony Optimization

- Ants are practically blind but they still manage to find their way to and from food. How do they do it?
- These observations inspired a new type of algorithm called **ant algorithms** (or ant systems)
- These algorithms are very new (Dorigo, 1996) and is still very much a research area
- Ant systems are a population based approach. In this respect it is similar to genetic algorithms
- There is a population of ants, with each ant finding a solution and then communicating with the other ants

Ant Colony Optimization

Double bridge experiments: different lengths



- Foraging behavior of real ants
- Blind ants communicate through stigmergy
- Leave pheromone trails to make a certain path more likely to be traversed by other ants

- The majority of ants follow the short path

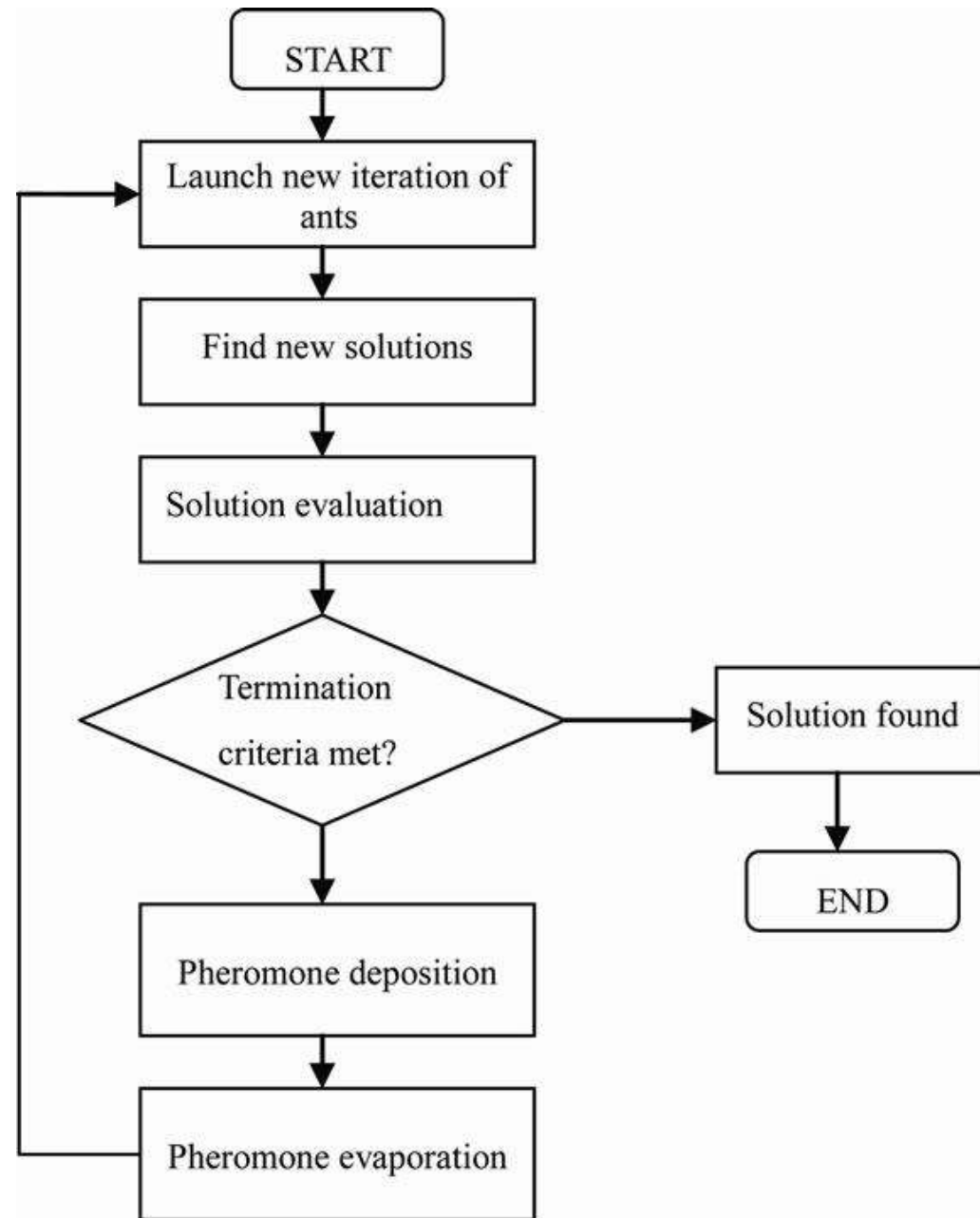
Ant Colony Optimization

- We are interested in exploring the search space, rather than simply plotting a route
- We need to allow the ants to explore paths and follow the best paths with some *probability* in proportion to the intensity of the pheromone trail
- We do not want them simply to follow the route with the highest amount of pheromone on it, else our search will quickly settle on a sub-optimal (and probably very sub-optimal) solution

Ant Colony Optimization

- The probability of an ant following a certain route is a function, not only of the pheromone intensity but also a function of what the ant can see (*visibility*)
- The pheromone trail must not build unbounded. Therefore, we need “evaporation”

Scheme of Ant Colony Optimization



Genetic Algorithm (bit details)

What is GA

- A **genetic algorithm** (or **GA**) is a search technique used in computing to find true or approximate solutions to optimization and search problems.
- Genetic algorithms are categorized as global search heuristics.
- Genetic algorithms are a particular class of evolutionary algorithms that use techniques inspired by evolutionary biology such as inheritance, mutation, selection, and crossover (also called recombination).

What is GA

- Genetic algorithms are implemented as a computer simulation in which a population of abstract representations (called chromosomes or the genotype or the genome) of candidate solutions (called individuals, creatures, or phenotypes) to an optimization problem evolves toward better solutions.
- Traditionally, solutions are represented in binary as strings of 0s and 1s, but other encodings are also possible.

What is GA

- The evolution usually starts from a population of randomly generated individuals and happens in generations.
- In each generation, the fitness of every individual in the population is evaluated, multiple individuals are selected from the current population (based on their fitness), and modified (recombined and possibly mutated) to form a new population.

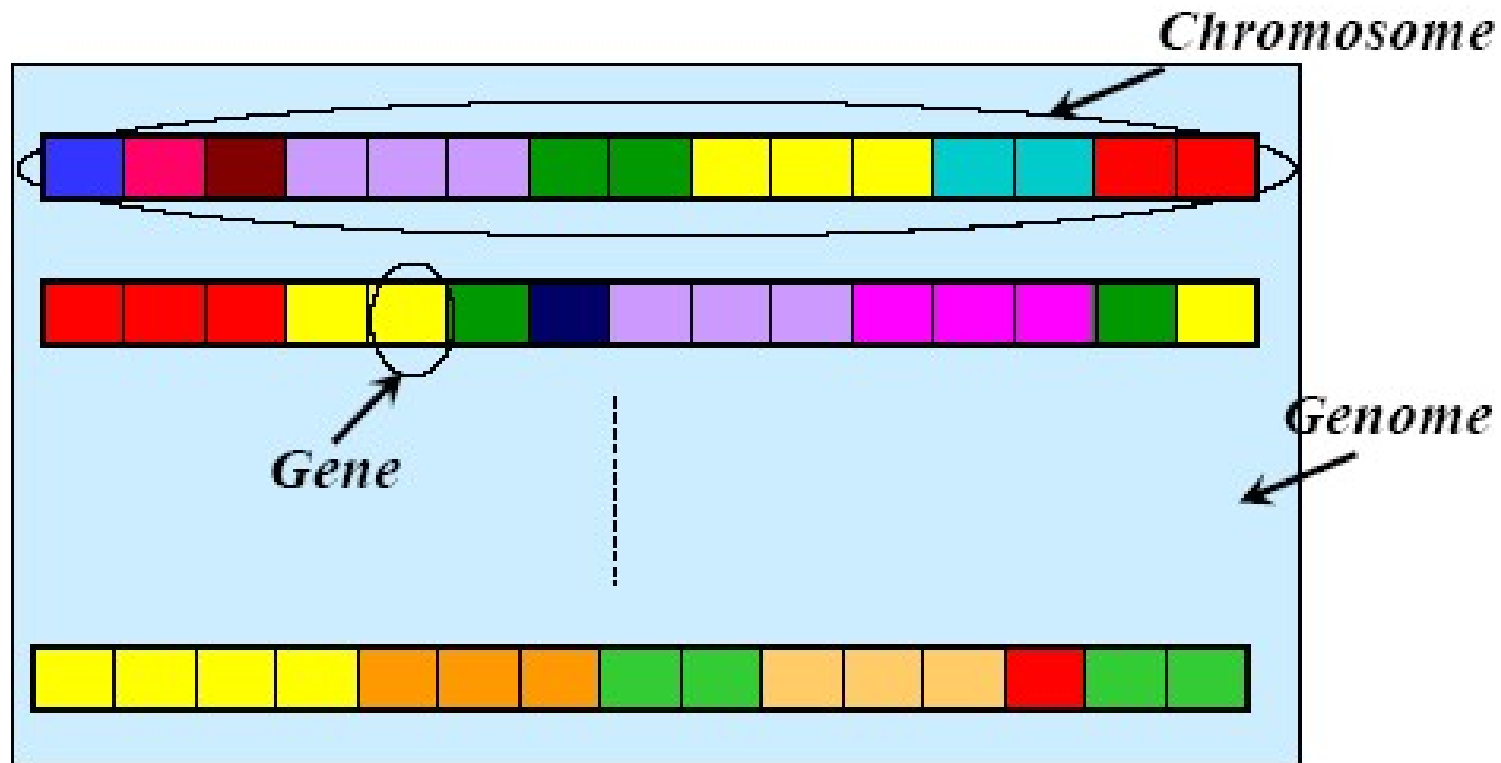
What is GA

- The new population is then used in the next iteration of the algorithm.
- Commonly, the algorithm terminates when either a maximum number of generations has been produced, or a satisfactory fitness level has been reached for the population.
- If the algorithm has terminated due to a maximum number of generations, a satisfactory solution may or may not have been reached.

Key terms

- **Individual** - Any possible solution
- **Population** - Group of all *individuals*
- **Search Space** - All possible solutions to the problem
- **Chromosome** - Blueprint for an *individual*
- **Trait** - Possible aspect (*features*) of an *individual*
- **Allele** - Possible settings of trait (black, blond, etc.)
- **Locus** - The position of a *gene* on the *chromosome*
- **Genome** - Collection of all *chromosomes* for an *individual*

Chromosome, Genes and Genomes



GA Requirements

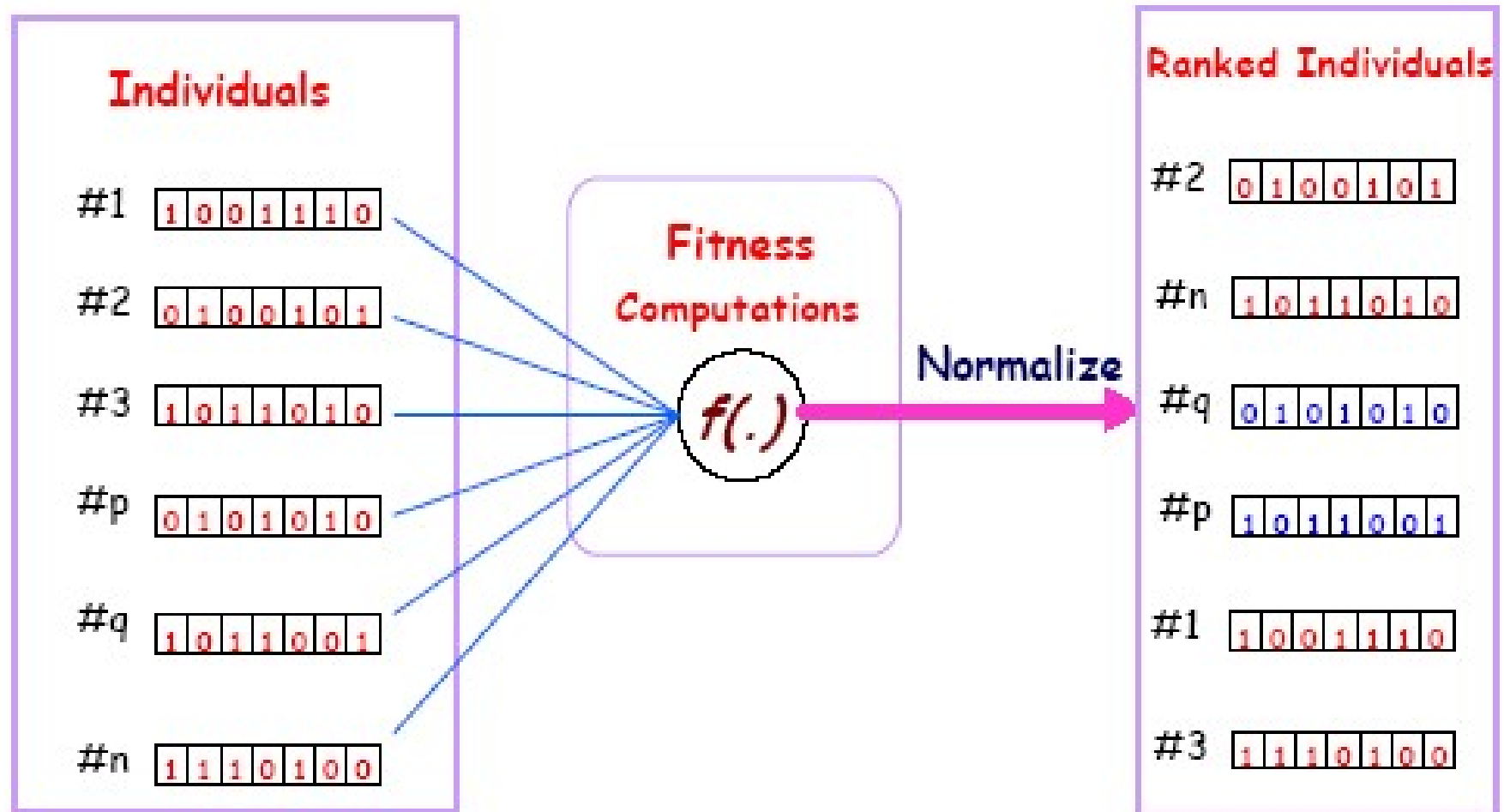
- A typical genetic algorithm requires two things to be defined:
 - a genetic representation of the solution domain, and
 - a fitness function to evaluate the solution domain.
- A standard representation of the solution is as an array of bits. Arrays of other types and structures can be used in essentially the same way.
- The main property that makes these genetic representations convenient is that their parts are easily aligned due to their fixed size, that facilitates simple crossover operation.
- Variable length representations may also be used, but crossover implementation is more complex in this case.

Representation

Chromosomes could be:

- Bit strings (0101 ... 1100)
- Real numbers (43.2 -33.1 ... 0.0 89.2)
- Permutations of element (E11 E3 E7 ... E1 E15)
- Lists of rules (R1 R2 R3 ... R22 R23)
- Program elements (genetic programming)
- ... any data structure ...

A fitness function



Basics of GA

- The most common type of genetic algorithm works like this:
- a population is created with a group of individuals created randomly.
- The individuals in the population are then evaluated.
- The evaluation function is provided by the programmer and gives the individuals a score based on how well they perform at the given task.
- Two individuals are then selected based on their fitness, the higher the fitness, the higher the chance of being selected.
- These individuals then "reproduce" to create one or more offspring, after which the offspring are mutated randomly.
- This continues until a suitable solution has been found or a certain number of generations have passed, depending on the needs of the programmer.

General Algorithm for GA

- **Initialization**
- Initially many individual solutions are randomly generated to form an initial population. The population size depends on the nature of the problem, but typically contains several hundreds or thousands of possible solutions.
- Traditionally, the population is generated randomly, covering the entire range of possible solutions (the *search space*).
- Occasionally, the solutions may be "seeded" in areas where optimal solutions are likely to be found.

General Algorithm for GA

- **Selection**

- During each successive generation, a proportion of the existing population is selected to breed a new generation.
- Individual solutions are selected through a *fitness-based* process, where fitter solutions (as measured by a fitness function) are typically more likely to be selected.
- Certain selection methods rate the fitness of each solution and preferentially select the best solutions. Other methods rate only a random sample of the population, as this process may be very time-consuming.
- Most functions are stochastic and designed so that a small proportion of less fit solutions are selected. This helps keep the diversity of the population large, preventing premature convergence on poor solutions. Popular and well-studied selection methods include roulette wheel selection and tournament selection.

General Algorithm for GA

- **Reproduction**

- The next step is to generate a second generation population of solutions from those selected through genetic operators:
crossover (also called recombination), and/or mutation.
- For each new solution to be produced, a pair of "parent" solutions is selected for breeding from the pool selected previously.
- By producing a "child" solution using the above methods of crossover and mutation, a new solution is created which typically shares many of the characteristics of its "parents". New parents are selected for each child, and the process continues until a new population of solutions of appropriate size is generated.

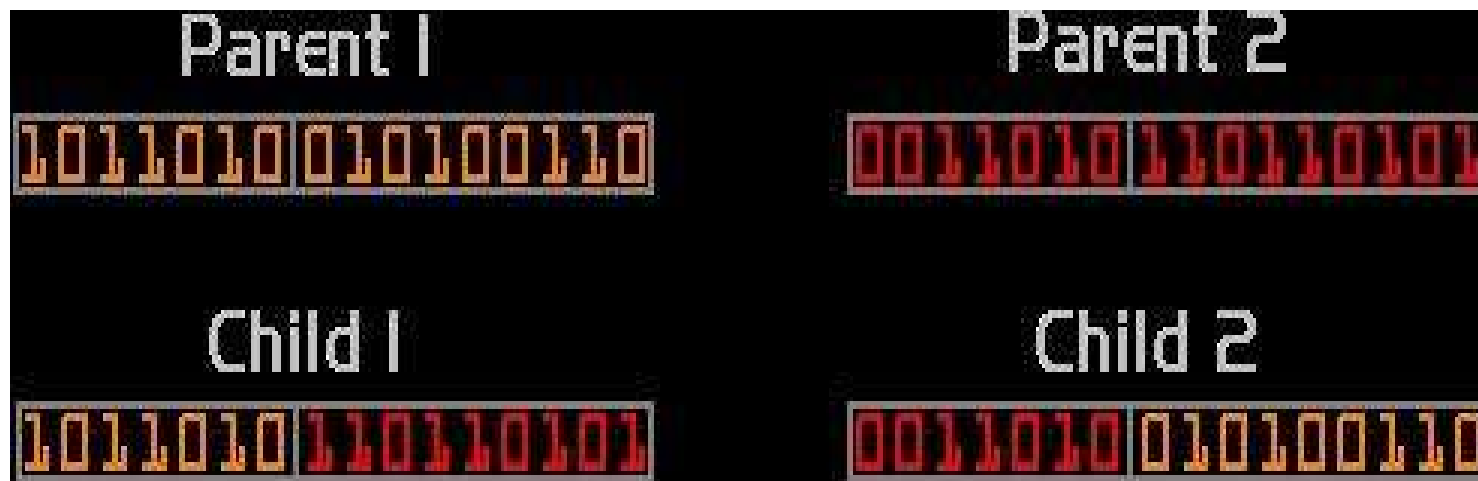
General Algorithm for GA

- These processes ultimately result in the next generation population of chromosomes that is different from the initial generation.
- Generally the average fitness will have increased by this procedure for the population, since only the best organisms from the first generation are selected for breeding, along with a small proportion of less fit solutions, for reasons already mentioned above.

Crossover

- the most common type is single point crossover. In single point crossover, you choose a locus at which you swap the remaining alleles from one parent to the other. This is complex and is best understood visually.
- As you can see, the children take one section of the chromosome from each parent.
- The point at which the chromosome is broken depends on the randomly selected crossover point.
- This particular method is called single point crossover because only one crossover point exists. Sometimes only child 1 or child 2 is created, but oftentimes both offspring are created and put into the new population.
- Crossover does not always occur, however. Sometimes, based on a set probability, no crossover occurs and the parents are copied directly to the new population. The probability of crossover occurring is usually 60% to 70%.

Crossover



Mutation

- After selection and crossover, you now have a new population full of individuals.
- Some are directly copied, and others are produced by crossover.
- In order to ensure that the individuals are not all exactly the same, you allow for a small chance of mutation.
- You loop through all the alleles of all the individuals, and if that allele is selected for mutation, you can either change it by a small amount or replace it with a new value. The probability of mutation is usually between 1 and 2 tenths of a percent.
- Mutation is fairly simple. You just change the selected alleles based on what you feel is necessary and move on. Mutation is, however, vital to ensuring genetic diversity within the population.

Mutation

Before: 1101101001101110
After: 1101101001101110

General Algorithm for GA

- **Termination**
- This generational process is repeated until a termination condition has been reached.
- Common terminating conditions are:
 - A solution is found that satisfies minimum criteria
 - Fixed number of generations reached
 - Allocated budget (computation time/money) reached
 - The highest ranking solution's fitness is reaching or has reached a plateau such that successive iterations no longer produce better results
 - Manual inspection
 - Any Combinations of the above

GA Pseudo-code

Choose initial population

Evaluate the fitness of each individual in the population

Repeat

- Select best-ranking individuals to reproduce

- Breed new generation through crossover and mutation (genetic operations) and give birth to offspring

- Evaluate the individual fitnesses of the offspring

- Replace worst ranked part of population with offspring

Until <terminating condition>

HOW ARE GENETIC ALGORITHMS DIFFERENT FROM TRADITIONAL METHODS?

In order for genetic algorithms to surpass their more traditional cousins in the quest for robustness, GAs must differ in some very fundamental ways. Genetic algorithms are different from more normal optimization and search procedures in four ways:

1. GAs work with a coding of the parameter set, not the parameters themselves.
2. GAs search from a population of points, not a single point.
3. GAs use payoff (objective function) information, not derivatives or other auxiliary knowledge.
4. GAs use probabilistic transition rules, not deterministic rules.

Case studies on Optimization...

(in the field of study)

Artificial Intelligence and Machine Learning

What is Artificial Intelligence?

In 1956, the computer scientist John McCarthy coined the term "Artificial Intelligence" (AI) to describe the study of intelligence by implementing its essential features on a computer.

Artificial intelligence (AI) is the intelligence exhibited by machines or software.

Artificial Intelligence: Introduction

Intelligence:

According to American Psychological Association (APA) **intelligence** refers to intellectual functioning. Expanding the definition we can describe intelligence as the ability to learn, understand or to deal with a new difficult situation. It is also the ability to apply one's knowledge to influence the surroundings.

It can be defined that an intelligent system is a system that emulates some aspects of intelligence exhibited by nature. These include learning, adaptability, robustness across problem domains, improving efficiency, information compression, extrapolated reasoning.

Artificial Intelligence:

Artificial intelligence (AI) is the science and engineering of making intelligent machines, especially intelligent computer programs. The ultimate goal is to make computer programs capable to solve problems similar to human beings.

Computational Intelligence:

Computational Intelligence is integrating the fields of Artificial Neural Networks, Genetic Algorithm, and Fuzzy Logic. It is the term formed by IEEE.

A few more terminologies

Soft Computing:

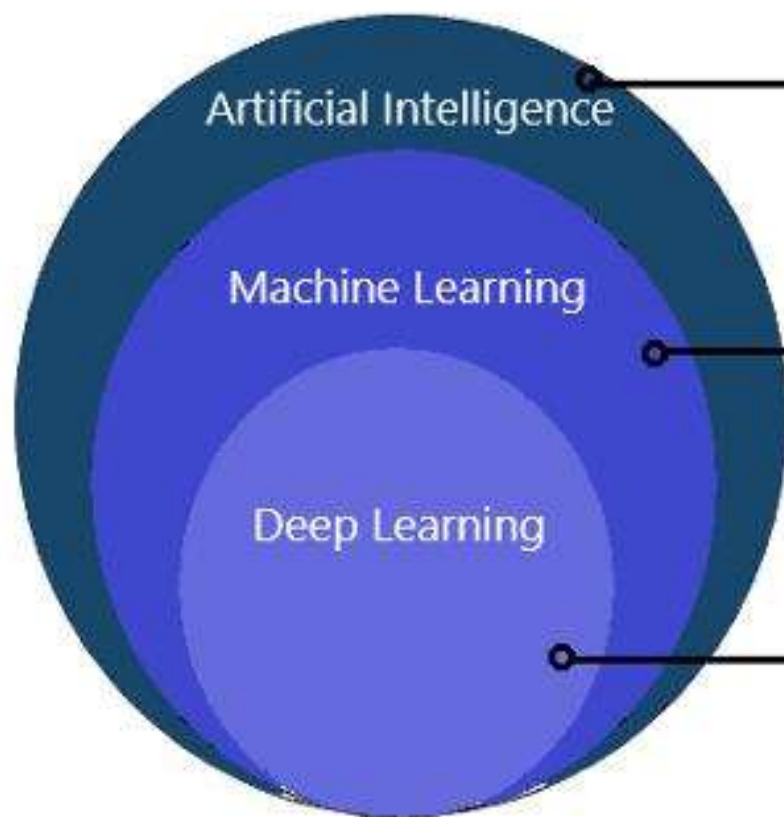
Soft computing is the collocation for the same fields as CI expanded with Rough Set Theory, Probabilistic Reasoning, Swarm Intelligence, and partly Chaos Theory.

Natural Computation:

Natural computation assembles mathematical methods, which are inspired by solution patterns from nature. Basically these are again Artificial Neural Networks, Evolutionary Approaches, Swarm Intelligence, etc.

Machine Learning:

Machine learning is based on Artificial Neural Networks, Support Vector Machines, Classification and Regression Trees, and some more similar methods. Important is that the resulting mathematical models are build by training from example data instead of being constructed analytically. This means that they are example based or empirical mathematical models instead of analytical ones.



Artificial Intelligence

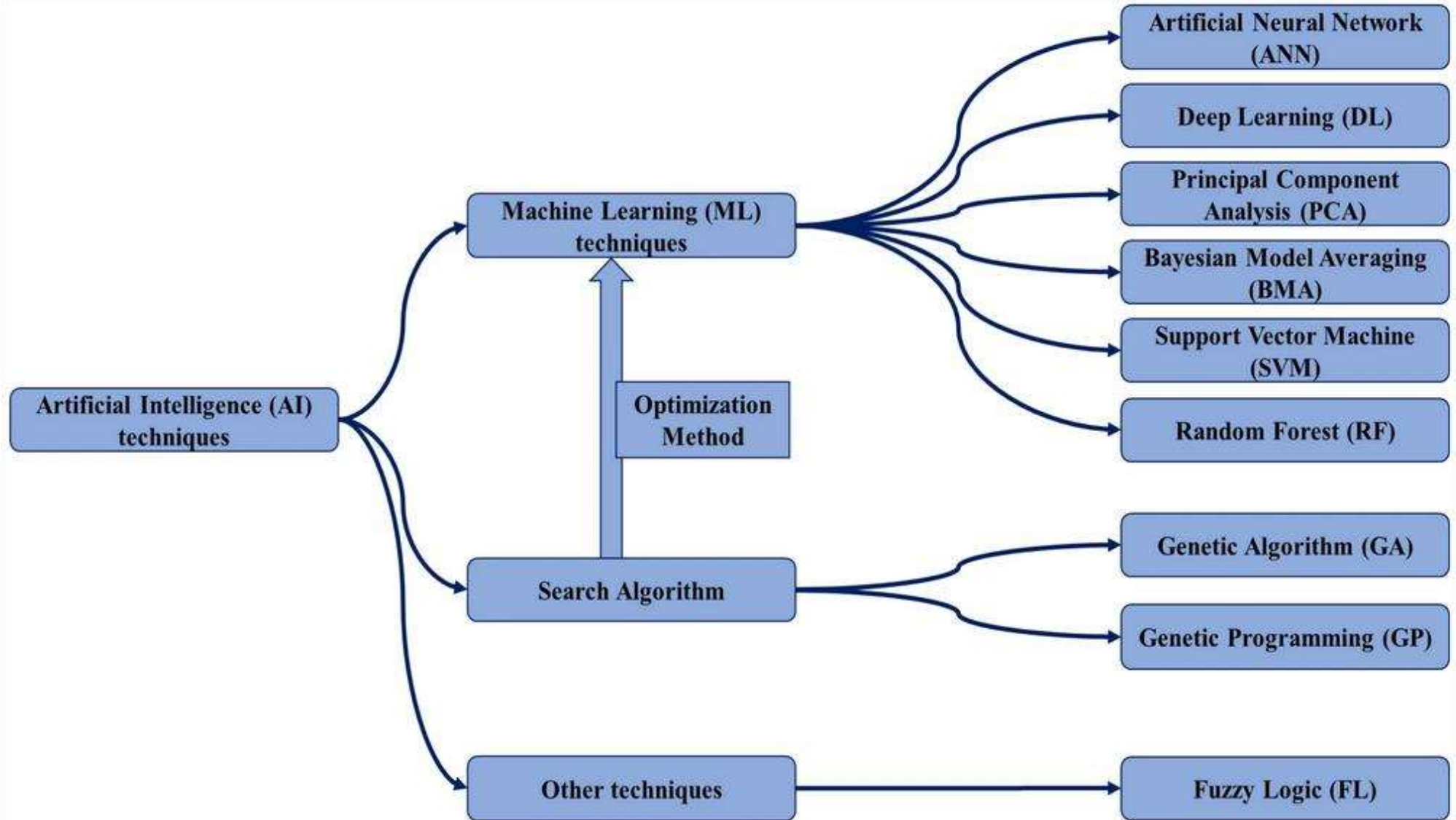
AI enables machines to think without any human intervention.

Machine Learning

Subset of AI that uses statistical learning algorithms that learn pattern in data over time

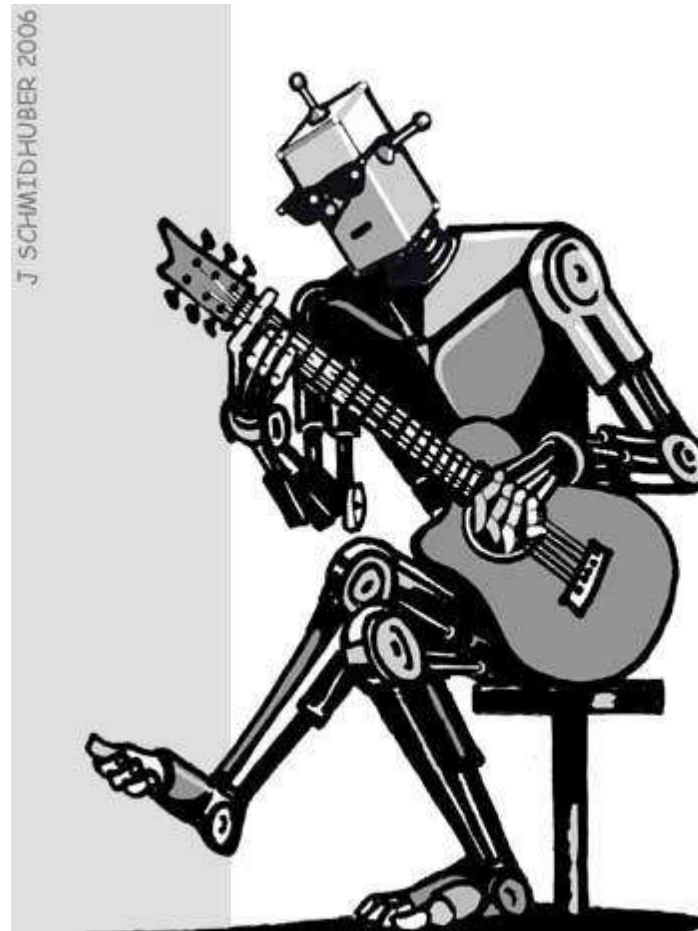
Deep Learning

Subset of ML that filters the data through multiple layers



Machine Learning

- To learn: *to get knowledge of by study, experience, or being taught.*
- Types of Learning
 - Supervised
 - Unsupervised



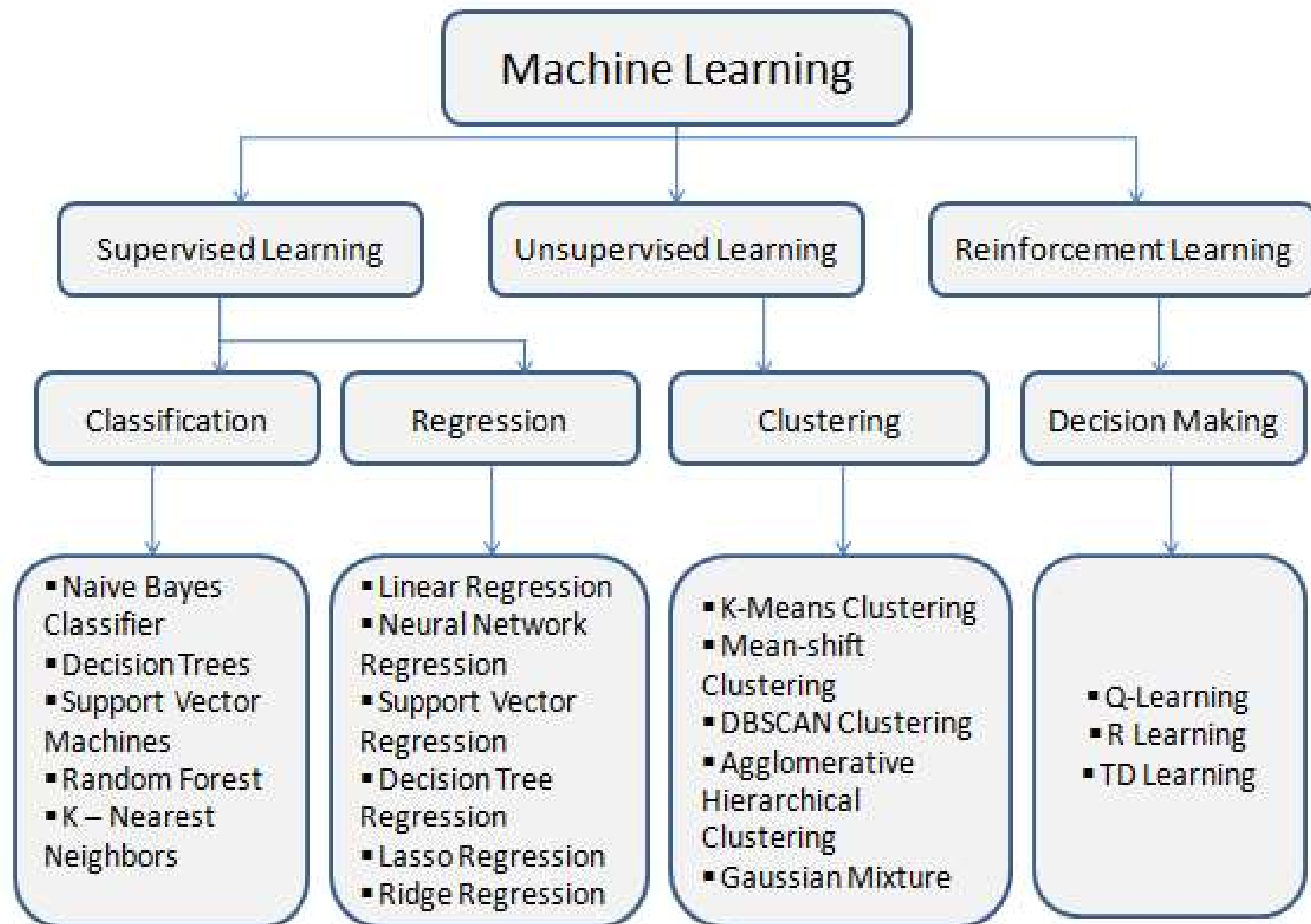
COGNITIVE ROBOTICS

Unsupervised Learning

- The model is not provided with the correct results during the training.
- Can be used to cluster the input data in classes on the basis of their statistical properties only.
- Cluster significance and labeling.
The labeling can be carried out even if the labels are only available for a small number of objects representative of the desired classes.

Supervised Learning

- Training data includes both the input and the desired results.
 - For some examples the correct results (targets) are known and are given in input to the model during the learning process.
 - The construction of a proper training, validation and test set is crucial.
 - These methods are usually fast and accurate.
- Have to be able to **generalize**: give the correct



Data and Information

- Data is a set of values of qualitative or quantitative variables.
- Data are simply facts or figures-- bits of information, but not information itself.
- When data are processed, interpreted, organized, structured or presented so as to make them meaningful or useful, they are called information.

Qualitative Data

*Qualitative data is extremely varied in nature. It includes virtually any information that can be captured that is **not numerical** in nature.*

- **In-Depth Interviews**

In-Depth Interviews include both individual interviews (e.g., one-on-one) as well as "group" interviews (including focus groups). The data can be recorded in a wide variety of ways including stenography, audio recording, video recording or written notes. In depth interviews differ from direct observation primarily in the nature of the interaction. In interviews it is assumed that there is a questioner and one or more interviewees. The purpose of the interview is to probe the ideas of the interviewees about the phenomenon of interest.

- **Direct Observation**

Direct observation is meant very broadly here. It differs from interviewing in that the observer does not actively query the respondent. It can include everything from field research where one lives in another context or culture for a period of time to photographs that illustrate some aspect of the phenomenon. The data can be recorded in many of the same ways as interviews (stenography, audio, video) and through pictures, photos or drawings (e.g., those courtroom drawings of witnesses are a form of direct observation).

- **Written Documents**

Usually this refers to existing documents (as opposed transcripts of interviews conducted for the research). It can include newspapers, magazines, books, websites, memos, transcripts of conversations, annual reports, and so on.

Quantitative Data

- **Quantitative** data are anything that **can be expressed as a number, or quantified**. Examples of quantitative data are scores on achievement tests, number of hours of study, or weight of a subject.
- There are two types of quantitative data, which is also referred to as numeric data: **continuous** and **discrete**. As a general rule, *counts* are discrete and *measurements* are continuous.
- **Discrete data** is a count that can't be made more precise. Typically it involves integers. For instance, the number of children (or adults, or pets) in your family is discrete data, because you are counting whole, indivisible entities: you can't have 2.5 kids, or 1.3 pets.
- **Continuous data**, is data that could be divided and reduced to finer and finer levels. For example, you can measure the height of your kids at progressively more precise scales—meters, centimeters, millimeters, and beyond—so height is continuous data.

Source of Data

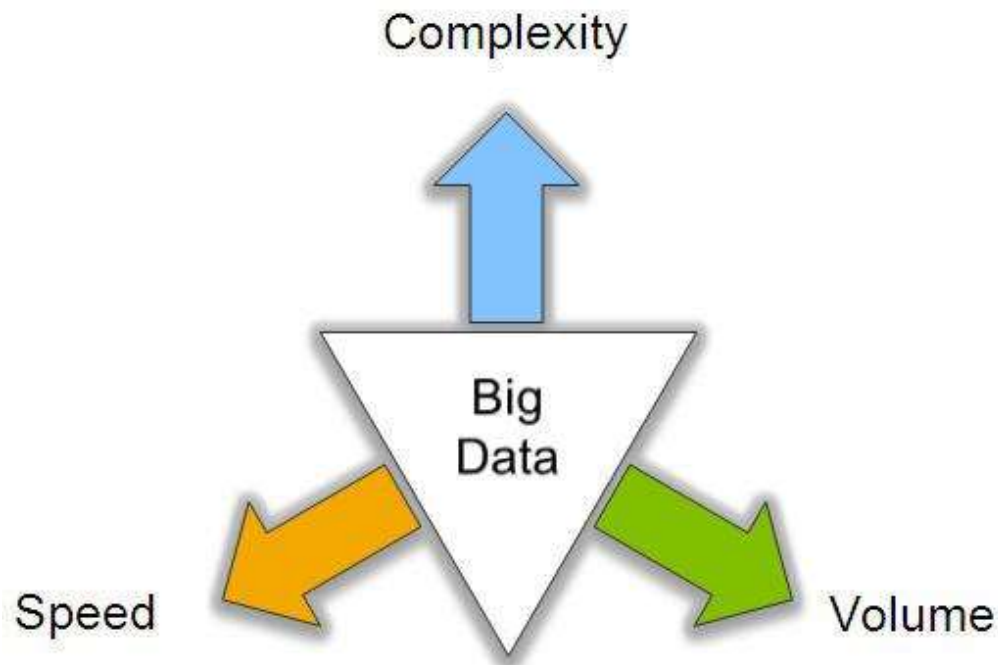
- Generated from experiments.
- Collected from published literatures.
- Industry data.

Data Analysis/Data Analytics

- **Analysis of data** is a process of inspecting, cleansing, transforming, and modeling data with the goal of discovering useful information, suggesting conclusions, and supporting decision-making.
- Data analysis has multiple facets and approaches, encompassing diverse techniques under a variety of names, in different business, science, and social science domains.
- Data analysis is a process for obtaining raw data and converting it into information useful for decision-making by users.

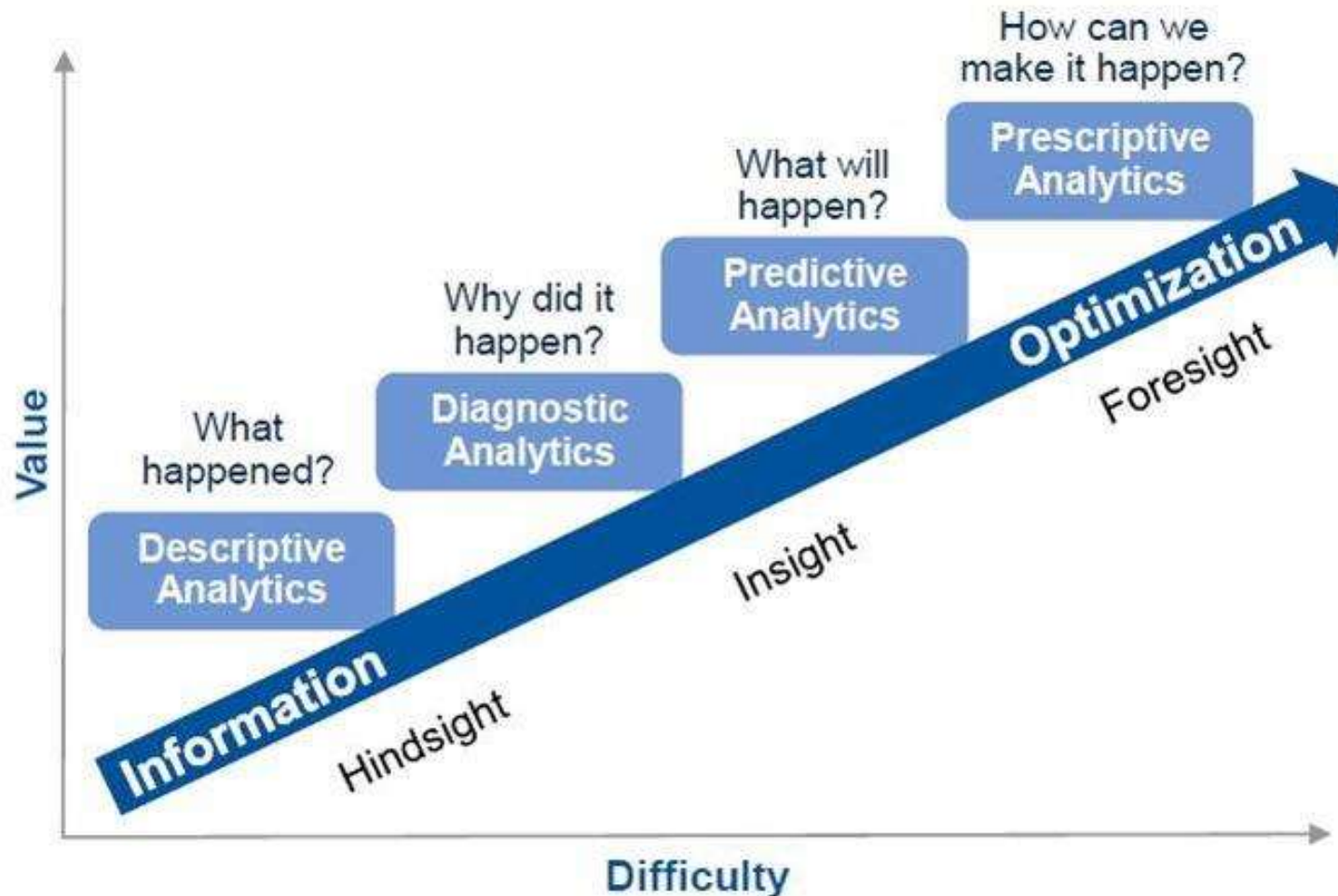
Big Data

- “***Big Data***” is data whose scale, diversity, and complexity require new architecture, techniques, algorithms, and analytics to manage it and extract value and hidden knowledge from it...



Data analytics

An area of data analytics dedicated to finding the best course of action for a given situation. Prescriptive analytics is related to both descriptive and predictive analytics.



Descriptive Analytics: Insight into the past

- Descriptive analysis or statistics does exactly what the name implies they “Describe”, or summarize raw data and make it something that is interpretable by humans.
- They are analytics that describe the past.
- The past refers to any point of time that an event has occurred, whether it is one minute ago, or one year ago.
- Descriptive analytics are useful because they allow us to learn from past behaviors, and understand how they might influence future outcomes.
- The vast majority of the statistics we use fall into this category. (Think basic arithmetic like sums, averages, percent changes). Usually, the underlying data is a count, or aggregate of a filtered column of data to which basic math is applied.

Diagnostic Analytics: Explaining the past

- Diagnostic analytics describes the techniques you will use to ask your data: Why did this happen?
- It's doing a deep-dive into your data to search for valuable insights. Descriptive analytics, the initial step in most companies' data analysis, is a simpler process that chronicles the facts of what has already happened.
- Diagnostic analytics takes it a step further to uncover the reasoning behind certain results.

Predictive Analytics: Understanding the future

- Predictive analytics has its roots in the ability to “Predict” what might happen. These analytics are about understanding the future.
- Predictive analytics provides actionable insights based on data.
- Predictive analytics provide estimates about the likelihood of a future outcome. It is important to remember that no statistical algorithm can “predict” the future with 100% certainty.
- These statistics forecast what might happen in the future.
- This is because the foundation of predictive analytics is based on probabilities.
- These statistics try to take the data that you have, and fill in the missing data with best guesses.
- They combine historical data to identify patterns in the data and apply statistical models and algorithms to capture relationships between various data sets.
- Predictive statistics and analytics are used to look into the future.
- Use Predictive Analytics any time you need to know something about the future, or fill in the information that you do not have.

Prescriptive Analytics: Advise on possible outcomes

- The relatively new field of prescriptive analytics allows users to “prescribe” a number of different possible actions to and guide them towards a solution.
- In a nut-shell, these analytics are all about providing advice.
- Prescriptive analytics attempt to quantify the effect of future decisions in order to advise on possible outcomes before the decisions are actually made.
- At their best, prescriptive analytics predicts not only what will happen, but also why it will happen providing recommendations regarding actions that will take advantage of the predictions.
- These analytics go beyond descriptive and predictive analytics by recommending one or more possible courses of action.
- Essentially they predict multiple futures and allow to assess a number of possible outcomes based upon their actions.
- Prescriptive analytics use a combination of techniques and tools such as various algorithms, machine learning and computational modelling procedures. These techniques are applied against input from many different data sets including historical and transactional data, real-time data feeds, and big data.
- Prescriptive analytics are relatively complex to administer but when implemented correctly, they can have a large impact on how decisions are made.

Predictive tools

- Regression
 - predict new values based on the past, inference
 - compute the new values for a dependent variable based on the values of one or more measured attributes
- Classification:
 - divide samples in classes
 - use a trained set of previously labeled data
- Clustering
 - partitioning of a data set into subsets (clusters) so that data in each subset ideally share some common characteristics
- Classification is in some way like the clustering but requires that the analyst know ahead of time how classes are defined.

ARTIFICIAL NEURAL NETWORKS

(bit details)

Neural Networks

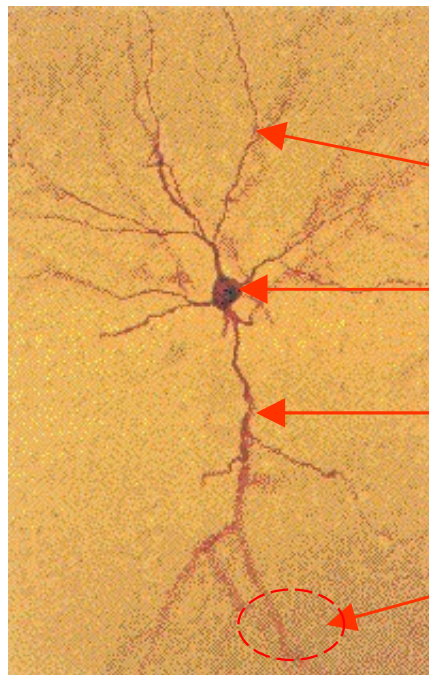
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- Artificial neural network (ANN) is a machine learning approach that models human brain and consists of a number of artificial neurons.
- Neuron in ANNs tend to have fewer connections than biological neurons.
- Each neuron in ANN receives a number of inputs.
- An activation function is applied to these inputs which results in activation level of neuron (output value of the neuron).
- Knowledge about the learning task is given in the form of examples called training examples.

Neural Network: Models that mimic functioning inside the human brain

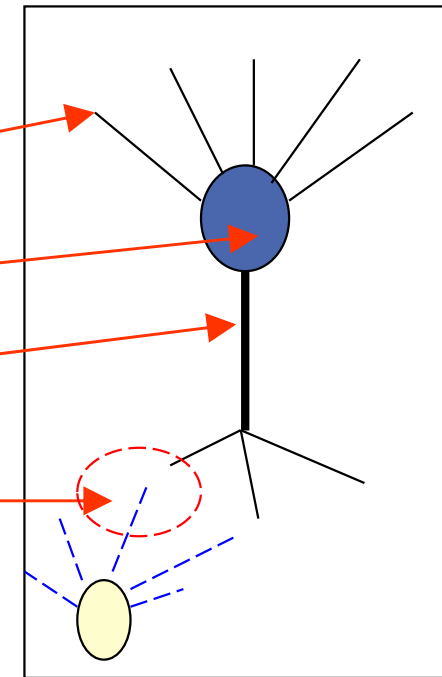
Most important functional unit in human brain – a class of cells called

NEURON



Hippocampal
Neurons

Dendrites
Cell Body
Axon
Synapse



Schematic

- **Dendrites** – Receive information
- **Cell Body** – Process information
- **Axon** – Carries processed information to other neurons
- **Synapse** – Junction between Axon end and Dendrites of other Neurons

- An Artificial Neural Network is specified by:
 - **neuron model**: the information processing unit of the NN,
 - **an architecture**: a set of neurons and links connecting neurons. Each link has a weight,
 - **a learning algorithm**: used for training the NN by modifying the weights in order to model a particular learning task correctly on the training examples.
- Need to construct a NN that is trained and generalizes well.
- It should behaves correctly on new instances of the learning task.

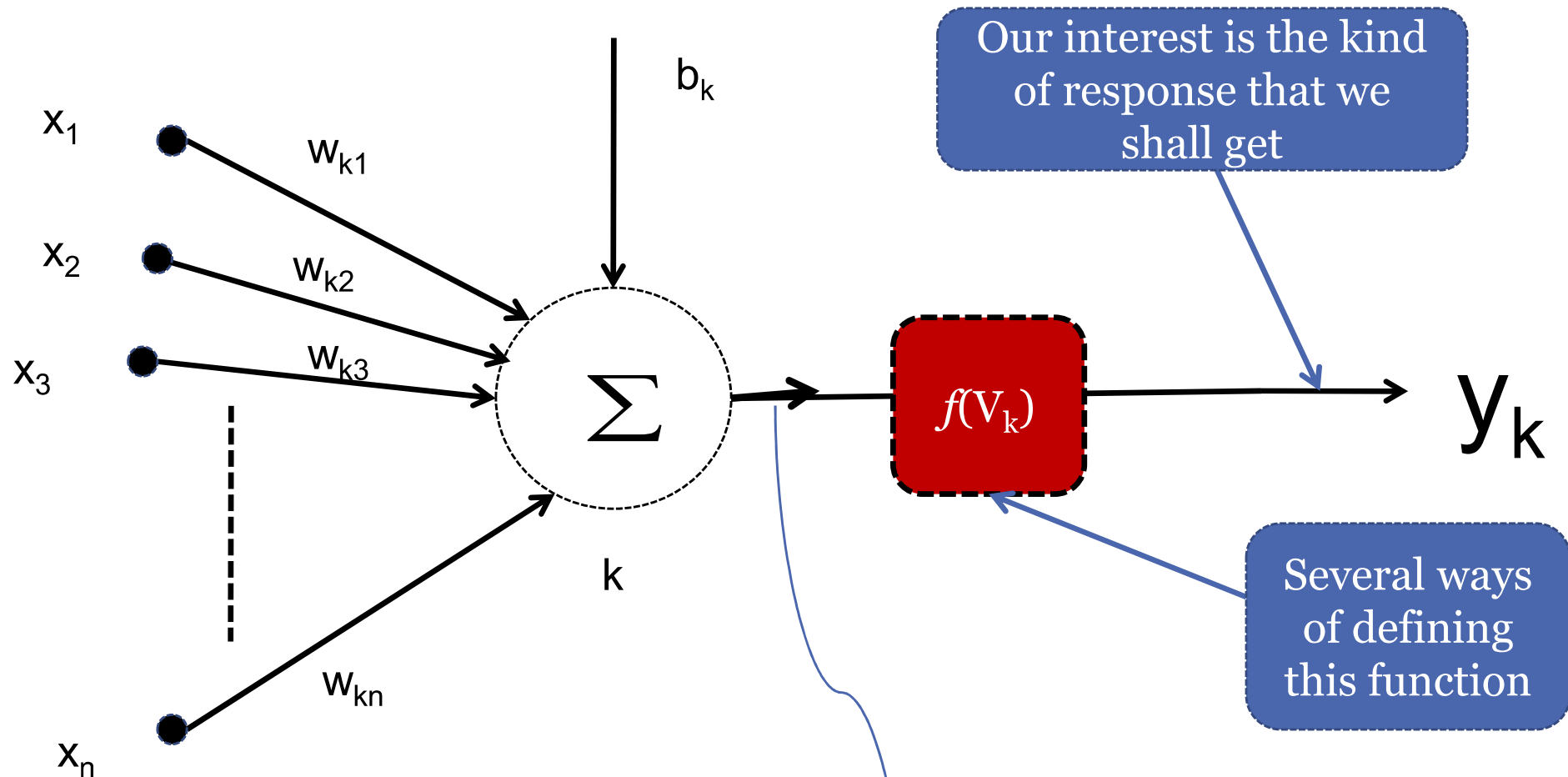
Neuron

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- The neuron is the basic information processing unit of a NN. It consists of:
 - 1 A set of **links**, describing the neuron inputs, with **weights** $W_1, W_2, \dots, W_n$
 - 2 An **adder** function (linear combiner) for computing the weighted sum of the inputs:
 - 3 **Activation function** for limiting the amplitude of the neuron output. Here 'b' denotes bias.

$$U_k = \sum_{j=1}^n x_j w_{kj}$$

$$y = \varphi(u + b)$$



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w_{kj} : connection strength from neuron j to neuron k

$$U_k = \sum_{j=1}^n x_j w_{kj}$$

$$V_k = U_k + b_k$$

Bias of a Neuron

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- The bias **b** has the effect of applying a **transformation** to the weighted sum **u**

$$v = u + b$$

- The bias is an external parameter of the neuron. It can be modeled by adding an extra input.
- **v** is called **induced field** of the neuron

$$v = \sum_{j=0}^n w_j x_j$$

$$w_0 = b$$

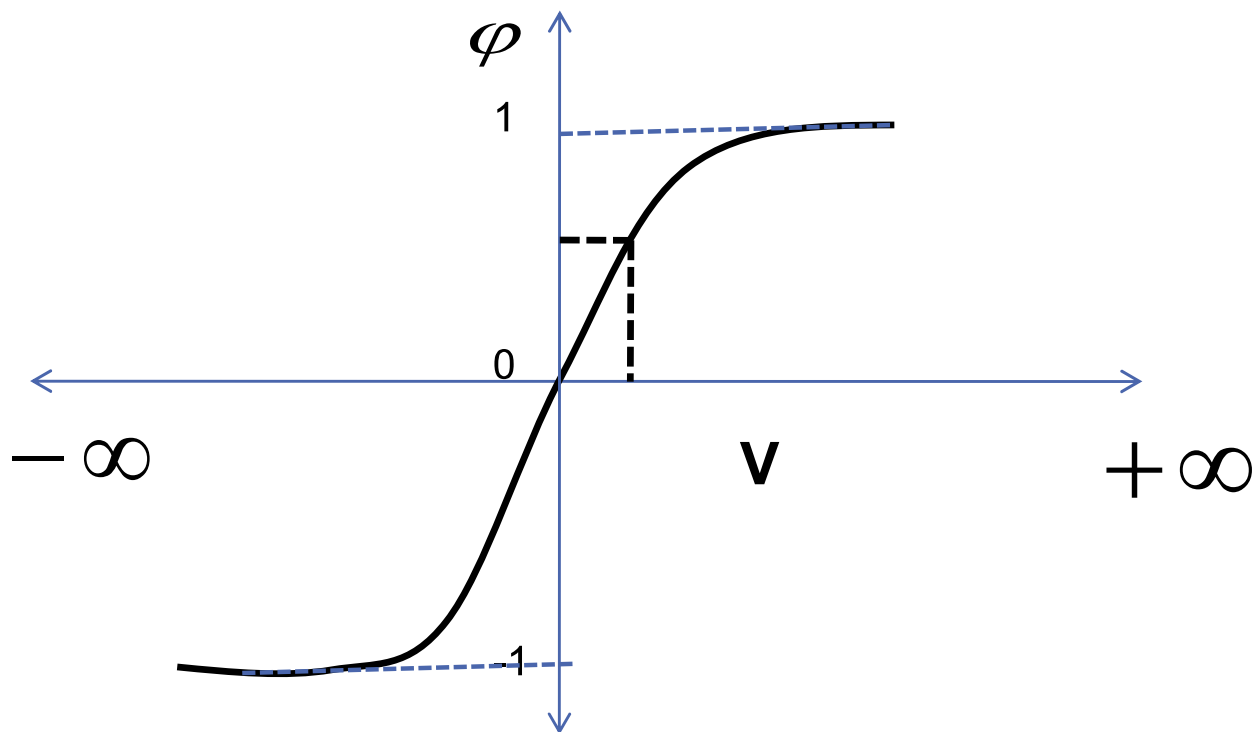
Neuron Models

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- The choice of activation function ϕ determines the neuron model.

Examples:

- step function:
$$\phi(v) = \begin{cases} a & \text{if } v < c \\ b & \text{if } v > c \end{cases}$$
- ramp function:
$$\phi(v) = \begin{cases} a & \text{if } v < c \\ b & \text{if } v > d \\ a + ((v - c)(b - a) / (d - c)) & \text{otherwise} \end{cases}$$
- sigmoid function with z,x,y parameters
$$\phi(v) = z + \frac{1}{1 + \exp(-xv + y)}$$
- Gaussian function:
$$\phi(v) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{v - \mu}{\sigma}\right)^2\right)$$



Non linear function b/w -1 & +1

$$\varphi(V) = \tanh(aV)$$

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Tanh

$$f(x) = (e^x - e^{-x}) / (e^x + e^{-x})$$

Deterministic NN: for a particular v , there is only 1

Neural Networks

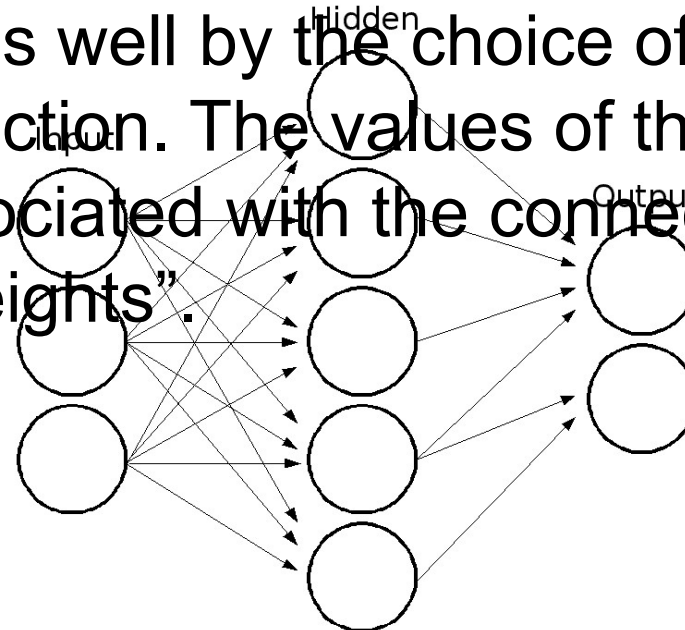
A Neural Network is usually structured into an input layer of neurons, one or more hidden layers and one output layer.

Neurons belonging to adjacent layers are usually fully connected and the various types and architectures are identified both by the different topologies adopted for the connections as well by the choice of the

The whole game of using NNs is in the fact that, in order for the network to yield appropriate outputs for given inputs, the weight must be set to suitable values.

activation function. The values of the functions associated with the connections are called “weights”.

The way this is obtained allows a further distinction among modes of operations.



Neural Networks: types

Feedforward: Single Layer Perceptron, MLP, ADALINE (Adaptive Linear Neuron), RBF

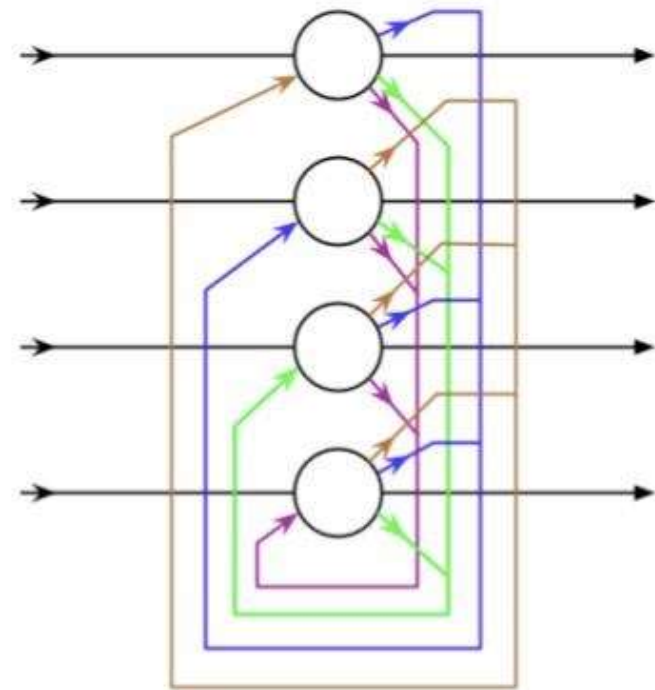
Self-Organized: SOM (Kohonen Maps)

Recurrent: Simple Recurrent Network, Hopfield Network.

Stochastic: Boltzmann machines, RBM.

Modular: Committee of Machines, ASNN (Associative Neural Networks), Ensembles.

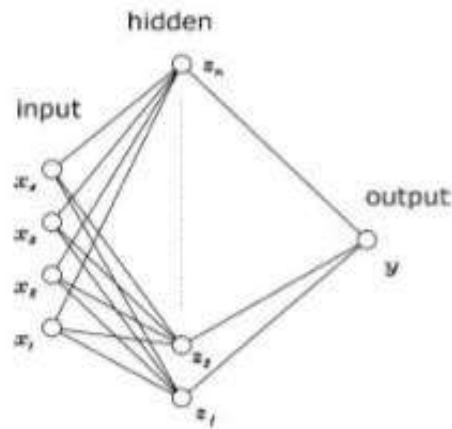
Others: Instantaneously Trained, Spiking (SNN), Dynamic, Cascades, NeuroFuzzy, PPS, GTM.



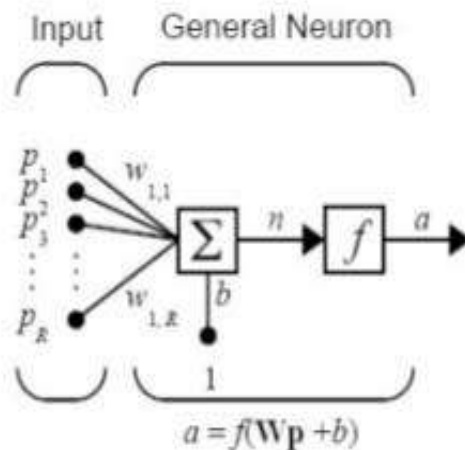
Multi Layer Perceptron (MLP) or feed-forward NN (FFNN)

The MLP is one of the most used supervised model: it consists of multiple layers of computational units, usually interconnected in a feed-forward way.

Each neuron in one layer has direct connections to all the neurons of the subsequent layer.

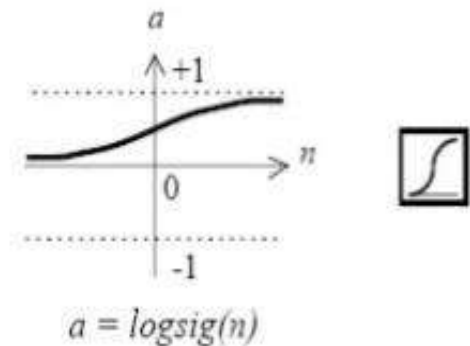


The architecture of a two layer MLP.



Where

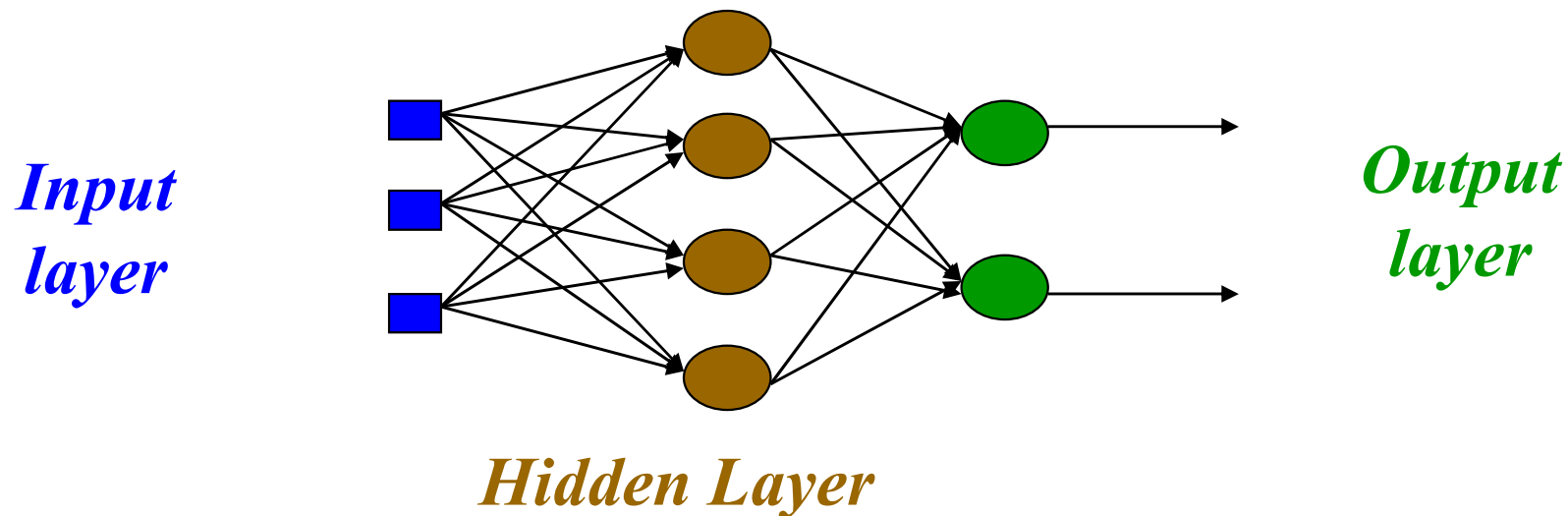
R = number of elements in input vector



Feed-forward NN (FFNN)

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- FFNN is a more general network architecture, where there are hidden layers between input and output layers.
- Hidden nodes do not directly receive inputs nor send outputs to the external environment.
- FFNNs overcome the limitation of single-layer NN.
- They can handle non-linearly separable learning tasks.

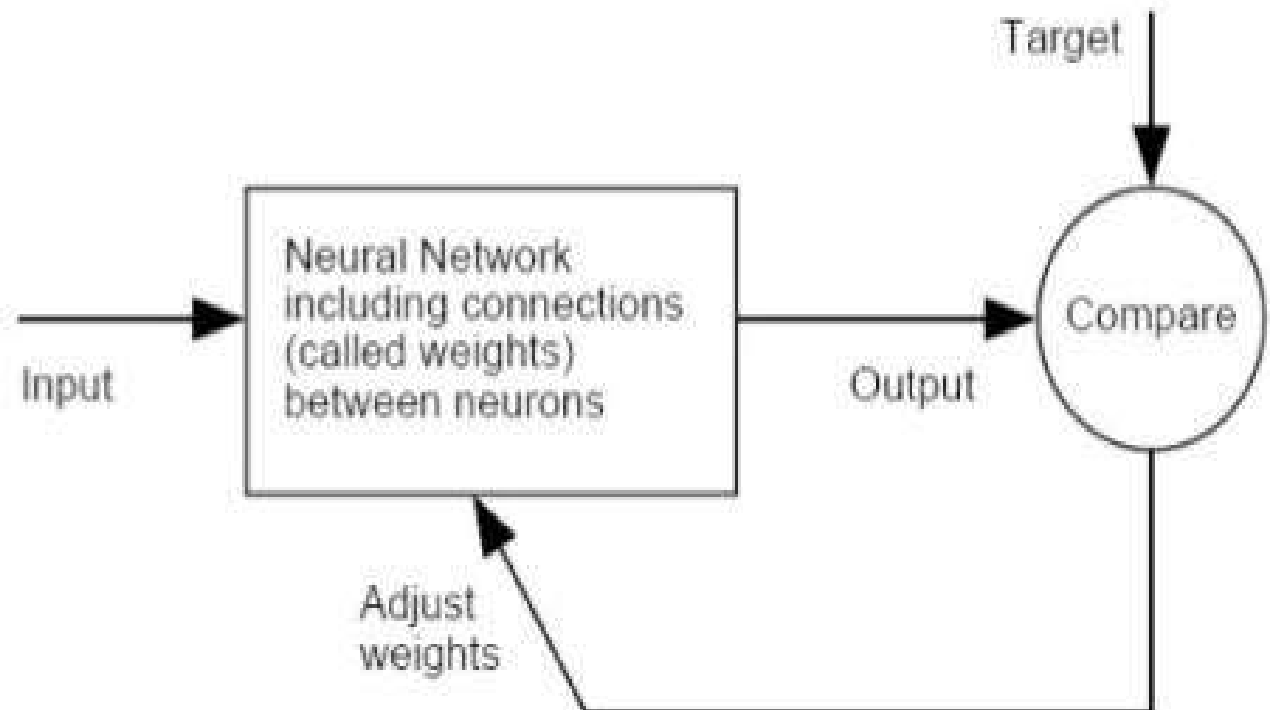


3-4-2 Network

Learning Process

- Back Propagation
 - the output values are compared with the target to compute the value of some predefined error function
 - the error is then feedback through the network
 - using this information, the algorithm adjusts the weights of each connection in order to reduce the value of the error function

After repeating this process for a sufficiently large number of training cycles, the network will usually converge.



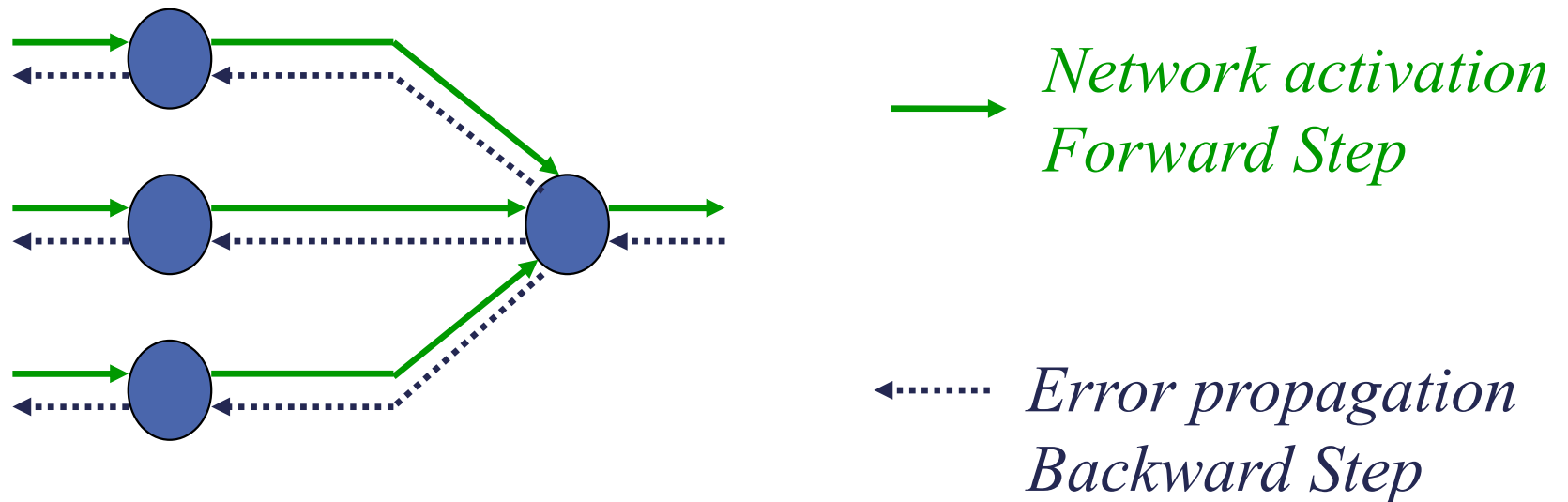
Training Algorithm: Backpropagation

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- The Backpropagation algorithm learns in the same way as single perceptron.
- It searches for weight values that minimize the total error of the network over the set of training examples (training set).
- Backpropagation consists of the repeated application of the following two passes:
 - **Forward pass:** In this step, the network is activated on one example and the error of (each neuron of) the output layer is computed.
 - **Backward pass:** In this step the network error is used for updating the weights. The error is propagated backwards from the output layer through the network layer by layer. This is done by recursively computing the local gradient of each neuron.

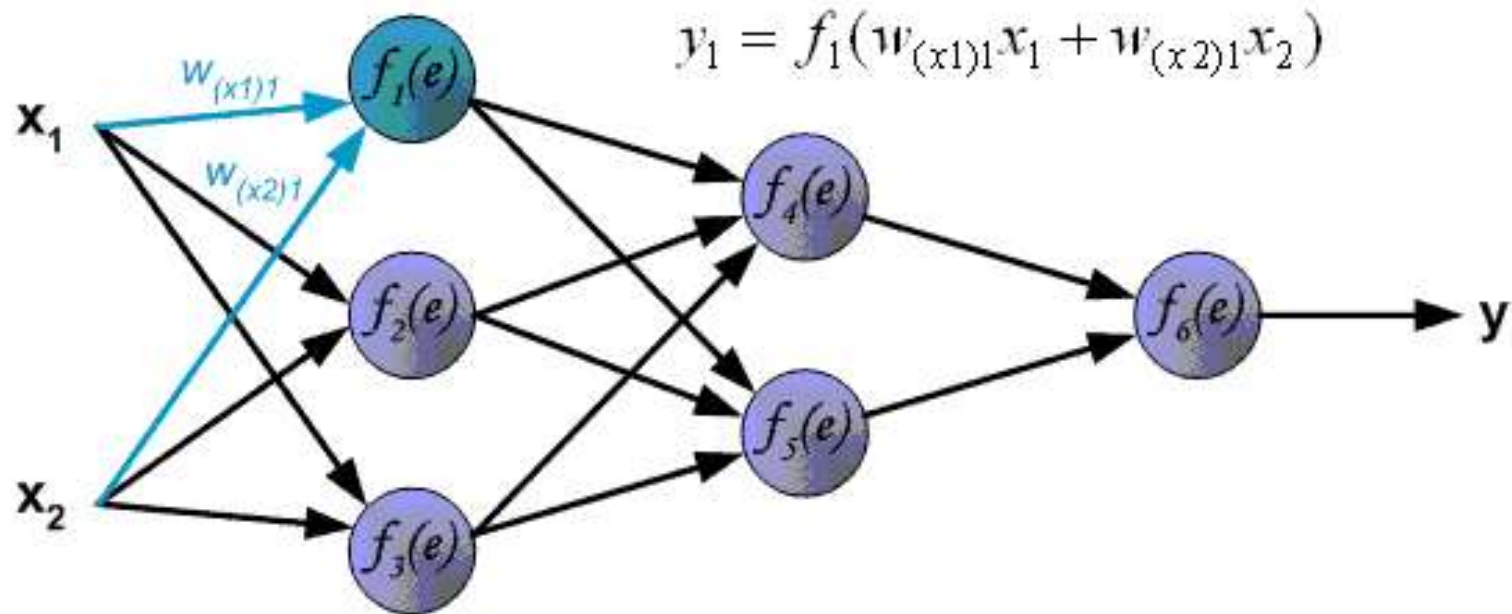
Backpropagation

- Back-propagation training algorithm



- Backpropagation adjusts the weights of the NN in order to minimize the network total mean squared error.

FP



Convergence Criterion

When to stop training the Network ?

Ideally – when we reach the **global minima** of the error surface

How do we know we have reached there ? We don't ...

Suggestion:

1. Stop if the decrease in total prediction error (since last cycle) is **small**.
2. Stop if the overall changes in the weights (since last cycle) are **small**.

Drawback:

Error keeps on decreasing. We get a **very good fit to training data**.

BUT ... The network thus obtained have **poor generalizing power** on **unseen data**

The phenomenon is also known as - **Over fitting** of the Training data

The network is said to **Memorize** the training data.

- so that when an X in training set is given,
the network faithfully produces the corresponding Y.

-However for X's which the network didn't see before, it predicts poorly.

Convergence Criterion

Modified Suggestion:

Partition the training data into **Training set** and **Validation set**

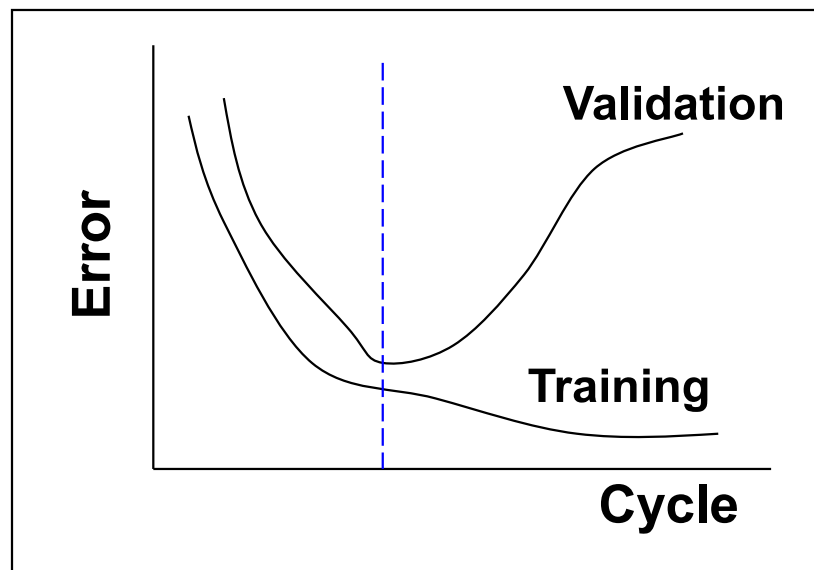
Use Training set - build the model

Validation set - test the performance of the model on unseen data

Typically as we have more and more training cycles

Error on Training set keeps on decreasing.

Error on Validation set keeps first decreases and then increases.



Stop training when the error on
Validation set starts increasing

Recurrent Network

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- **FFNN** is acyclic where data passes from input to the output nodes and not vice versa.
 - Once the FFNN is trained, its state is fixed and does not alter as new data is presented to it. It does not have memory.
- **Recurrent network** can have connections that go backward from output to input nodes and models dynamic systems.
 - In this way, a recurrent network's internal state can be altered as sets of input data are presented. It can be said to have memory.
 - It is useful in solving problems where the solution depends not just on the current inputs but on all previous inputs.
- **Applications**
 - predict stock market price,
 - weather forecast

Recurrent network

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Motivation

- Need for Systems which can process time dependant data.

Especially for applications (like weather forecast)
which involves prediction based on the past.

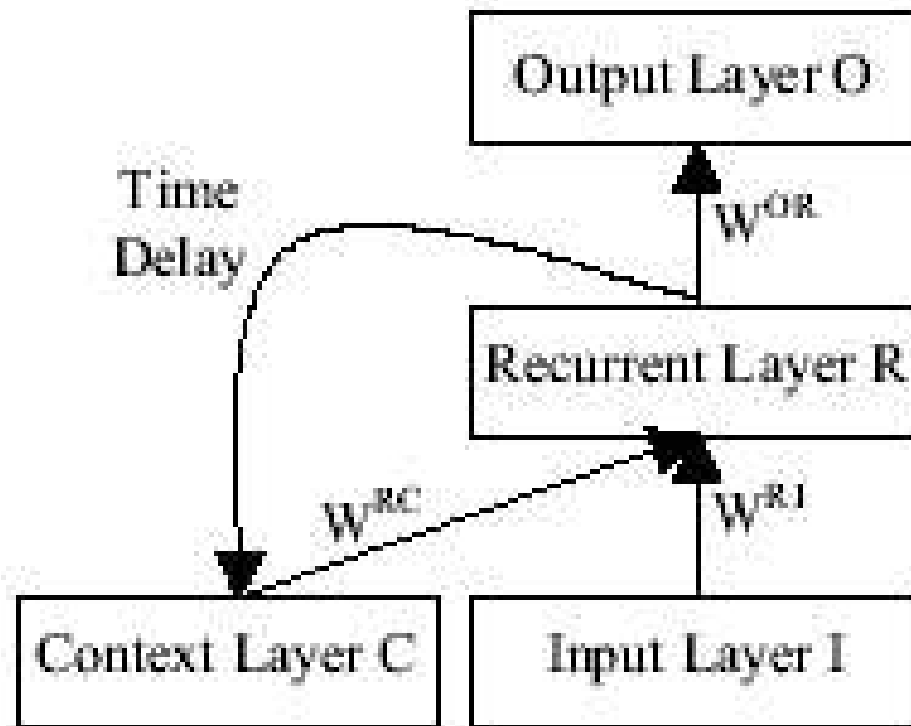
Introduction to RNN

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- Feedforward neural networks are unable to process data with time dependent information.
- In RNN (Recurrent neural network) the hidden or output units can be provided with an extended input that is composed of current input activities together with the activities from the previous time step.
- RTRL(Real time recurrent learning algorithm) used for training RNN are based on gradient minimization of error.

Simple Recurrent Network(SRN)

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Simple Recurrent network (SRN)

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- It is multi layer perceptron with feedback connection.
- It consists of **input , output , recurrent and context layer**.
- The recurrent layer is fed through recurrent connections by activities of the context layer.
- Context unit hold activities of the recurrent layer from previous time step.
- The units of the input layer I and recurrent layer R are fully connected through weight W^{RI} and output layer O by weight W^{OR}

NN DESIGN ISSUES

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- Data representation
- Network Topology
- Network Parameters
- Training
- Validation

Data Representation

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- Attributes of different types may have different ranges of values which affect the training process.
- Normalization may be used, like the following one which scales each attribute to assume values between 0 and 1.

$$x_n = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- Normalization between any arbitrary points 'a' and 'b'

$$x_n = a + \frac{(x - x_{\min})(b - a)}{(x_{\max} - x_{\min})}$$

Network Topology

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- The number of layers and neurons depend on the specific task.
- In practice this issue is solved by trial and error.
- Two types of adaptive algorithms can be used:
 - start from a large network and successively remove some neurons and links until network performance degrades.
 - begin with a small network and introduce new neurons until performance is satisfactory.

Network parameters

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- How are the weights initialized?
- How is the learning rate chosen?
- How many hidden layers and how many neurons?
- How many examples in the training set?

Case studies on Machine Learning ... (in the field of study)